

# What Held-Out Accuracy Predicts About Deploying Intrusion Detection in O-RAN

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**ABSTRACT** Intrusion detectors for 5G and O-RAN traffic are often reported above 99% accuracy on a random split of one corpus. We ask what such a number predicts about deployment, using NetsLab-5GORAN-IDD, 5G-NIDD, six architectures, two input-blind baselines, and a ladder of protocols. A random split raises radio-layer macro- $F_1$  by 0.13 (95% interval -0.01 to 0.27; significant for two sequence models), and a nearest-neighbor lookup gains 0.15, consistent with models recognizing capture sessions. The radio capture is ordered by scenario, so a time-disjoint split holds out whole attack categories. With coverage fixed, time order is indistinguishable from session variation, and a held-out benign session draws a false positive rate from 0.00 to 0.98. In 5G-NIDD, 59% of benign flows copy flood records labeled as attacks elsewhere, capping balanced accuracy at 0.766. The dataset authors' 99.9% accuracy reproduces, but it rests on two record-position fields that separate the conflicting records; without them it falls to 76.9%. Transferred to 5G-NIDD flows with consistent labels, detectors reach balanced accuracy 0.61 to 0.78, below the target's own reference, and none passes a deployability test that the published pipeline passes on its own split. A third corpus shows the same pattern. At a prevalence of 0.002, no threshold that keeps recall above 10% lifts operational precision beyond 0.063, and ten benign radio sessions cannot bound the radio alert burden. Averaging precision over folds inflates the best-to-worst ratio between architectures from 1.45 to 16.0. We close with reporting requirements for deployability claims.

**INDEX TERMS** Dataset shift, evaluation methodology, false positive rate, intrusion detection, label quality, O-RAN, RAN intelligent controller, xApp.

## I. INTRODUCTION

Open Radio Access Network (O-RAN) architectures dis-aggregate the base station into radio, distributed, and central units connected by open interfaces, and add a RAN Intelligent Controller (RIC) that hosts third-party control applications [1]–[3]. The near-real-time RIC runs close to the cells it controls [1], [4], so it observes the telemetry of every device those cells serve. Compromised devices, many of them IoT devices, are a documented source of volumetric attacks [5], [6]. That makes a detection application (xApp) inside the RIC control loop an attractive defense: platforms for building xApps exist [7], [8], and detectors that run in or beside the O-RAN control loop have been built [9], [10].

Detectors for 5G and O-RAN traffic are usually evaluated on a random held-out split of one dataset, and the reported accuracy or  $F_1$  is commonly above 99%. On 5G-NIDD, one of the two corpora we use, the dataset's authors report binary accuracy of 99.85% to 99.95% for four of five classifiers [11], and later work reports weighted  $F_1$  of up to 99.95% [12]. Three properties decide whether such a detector is deployable, and a held-out split of the training distribution measures none of them. The first is generalization to a deployment

that did not supply the training data; intrusion detectors are known to transfer poorly between independently collected datasets [13], [14]. The second is alert burden: a false positive rate that looks small on a balanced benchmark becomes a large alert volume once it is multiplied by production flow rates and read against a realistic attack base rate [15], [16]. The third is latency against the near-real-time control loop, for which the O-RAN E2 specification requires support of latencies from 10 ms to 1 s [17]. A benchmark score measures how well a detector separates the records of one corpus; deployment asks how it behaves on traffic it was not trained on, at a realistic base rate, inside a control loop. We treat the first as evidence about the second only where an experiment shows that it is.

Each of these critiques has been made before. The base-rate argument is Axelsson's [15], the closed-world critique is Sommer and Paxson's [16], and evaluation pitfalls in security machine learning have been catalogued by Arp et al. [18], with temporal and spatial bias formalized by Pendlebury et al. [19]. Transfer between the two corpora we use has been measured by Abraheem and Edhirig [20], and inference inside a real near-real-time RIC has been measured by Obiuevewi

et al. [21]. Our contribution is not a new detector and not the observation that transfer degrades. It is a single, controlled evaluation of one detector family in which these effects are measured together, with trivial baselines evaluated on every corpus, so that their relative sizes can be compared and the claims that commonly rest on them can be tested. Specifically:

- We measure a ladder of evaluation protocols on one pipeline, with six architectures and two input-blind baselines. A model-free nearest-neighbor probe and two sequence models point to the recognition of capture sessions as the source of the random-split gain (Section VI-A), and a split that fixes category coverage shows that the radio corpus supports no drift estimate, while the benign session in the test set sets the false positive rate (Section VI-B).
- We audit the target corpus and show that half of 5G-NIDD's flows sit on records that carry both labels, and that a published 99.9% result on it rests on record-position fields; we take that published pipeline down the same ladder (Section VI-C).
- We separate prior shift, label conflicts, and transfer failure in the cross-corpus gap with a prevalence-invariant metric, a difference-in-differences test against the majority baseline, references measured inside the target, and evaluation with and without the conflicting records, and we repeat the transfer on a third corpus from a different site and exporter (Sections VI-D, VI-E and VI-F).
- We report operational precision and alert volume from pooled confusion counts in explicit units with cluster-bootstrap intervals, and show that averaging precision over evaluation folds inflates the apparent difference between architectures by an order of magnitude (Section VI-G).
- We show which calibration methods can change the achievable operating points, that none repairs them here, and that label-free estimation of the target prior fails when class-conditional distributions shift (Section VI-H).
- We decompose decision latency on one host for three implementations, including a compiled runtime (Section VI-I). We then combine generalization, alert burden, and latency into a deployability test, show how its verdict depends on its bounds, and derive reporting requirements (Sections VII and X).

All latency figures are emulated, with no RIC in the path. The three corpora come from three different flow exporters, so every cross-corpus gap mixes a change of deployment with a change of tooling. Section VI-E measures how much that matters for the first two. The notation used throughout is summarized in Table 1.

## II. BACKGROUND AND RELATED WORK

This section places the study in four bodies of work: the RIC as a platform for detection, O-RAN security, datasets and learning methods for network intrusion detection, and

TABLE 1. Notation and abbreviations.

| Symbol                             | Meaning                                                     |
|------------------------------------|-------------------------------------------------------------|
| $\mathcal{D}_A, \mathcal{D}_B$     | source corpus (NetsLab-5GORAN-IDD); target corpus (5G-NIDD) |
| $\mathcal{D}_C$                    | third corpus, a second target (5G core datasets of [22])    |
| $\tau$                             | decision threshold on the attack score $\hat{p}$            |
| $R(\tau), \text{FPR}(\tau)$        | recall and false positive rate at $\tau$                    |
| $\pi$                              | operational attack prevalence (declared, not measured)      |
| $\pi_{\text{tr}}$                  | attack prevalence of the training data                      |
| $\lambda_b$                        | benign arrival rate (flows or windows per hour)             |
| $A(\tau)$                          | false alerts per hour at $\tau$                             |
| $\text{PPV}(\tau)$                 | operational precision at prevalence $\pi$                   |
| $\Delta_{BA}, \Delta_{F1}$         | source minus target balanced accuracy, macro- $F_1$         |
| $\bar{W}_1$                        | mean first-order Wasserstein distance over features         |
| $L, q_p(L)$                        | decision latency and its $p$ -th quantile                   |
| $B, \varepsilon$                   | latency budget and tolerated violation rate                 |
| $\delta, A_{\max}, r_{\min}, \rho$ | bounds of the deployability test, Eq. (6)                   |
| $\delta_m$                         | deadline margin in Algorithm 1 (design, not evaluated)      |
| BA                                 | balanced accuracy, the mean of TPR and TNR                  |
| LR, DT, RF                         | logistic regression, decision tree, random forest           |
| XGB, HGB, MLP                      | XGBoost, histogram gradient boosting, multilayer perceptron |
| IF, AE, GRU                        | isolation forest, autoencoder, gated recurrent unit         |
| KPM, RC                            | E2 service models for measurements and RAN control          |
| UE                                 | user equipment                                              |

the literature on evaluation practice and dataset shift that the paper applies.

### A. THE NEAR-REAL-TIME RIC AS A DETECTION PLATFORM

xApps subscribe to E2 nodes through service models: E2SM-KPM delivers periodic key performance measurements and E2SM-RC exposes control actions [1]. Platforms such as OpenRAN Gym [23], SCOPE [24], Colo-RAN [7], and FlexRIC [8], together with open radio stacks [25], [26], make experimental xApps practical. E2 carries aggregated measurements rather than packets, which matters for the data path of a flow-level detector (Section VII-A).

### B. SECURITY OF OPEN RAN

The O-RAN Alliance's threat model [27] and independent analyses [28], [29] identify the RIC and third-party xApps as a new attack surface, and security mechanisms across O-RAN interfaces have been implemented and measured [30]. On the detection side, 5G-Spector runs layer-3 attack detection as an O-RAN compliant service [9], Det-RAN detects attacks from cross-layer data in real time [10], and Fard et al. study the fusion of radio telemetry and flow records on the same source corpus we use [31]. These works propose or evaluate detectors on one deployment; our question is what evidence would show that such a detector works elsewhere.

### C. MACHINE LEARNING FOR NETWORK INTRUSION DETECTION

Many IoT-oriented benchmark corpora exist [32]–[36]. Cellular corpora are rarer: 5G-NIDD was captured on a functioning 5G test network [37], [38], and NetsLab-5GORAN-IDD pairs flow records with radio telemetry from an O-RAN testbed [39]. On 5G-NIDD, the dataset's authors report binary accuracy of 99.85% to 99.95% for four of five classifiers on a random 70/30 split [11], and later work reports weighted

$F_1$  of up to 99.95% [12]. Standardized NetFlow feature sets make cross-corpus experiments possible [40], [41]. The quality of benchmark corpora is itself a source of error: labeling and flow-construction faults have been documented in CIC-IDS2017 [42], and design flaws that let a classifier score well without learning the attack recur across NIDS benchmarks [43].

### D. EVALUATION PRACTICE AND DATASET SHIFT

Intrusion detection results are often over-optimistic because of closed-world evaluation [16], base-rate neglect [15], sampling bias and leakage [18], [44], and temporal or spatial bias in the split [19]; the gap between reported and operational effectiveness is examined across security domains in [45], and industrial intrusion detection shows the same pattern when detectors are re-evaluated on data they were not tuned on [46]. Cross-dataset generalization of such detectors is poor [13], [14], [47]. Closest to this paper, Abraheem and Edhirig transfer XGBoost between our two corpora on 15 harmonized features and report zero-shot balanced accuracy per attack family of 0.50 to 0.56 from NetsLab-5GORAN-IDD to 5G-NIDD and 0.52 to 0.83 in the reverse direction [20]. Our transfer results agree with theirs in direction and size. What this paper adds is the protocol ladder with input-blind baselines, references measured inside the target corpus, a published pipeline taken down the same ladder, an audit of the target's labels, and the operational and latency criteria. The dataset-shift literature supplies tools this paper uses: label-shift estimation from unlabeled data [48], [49], calibration and its behavior under shift [50]–[52], shift detection [53], and covariance alignment for domain adaptation [54]. For comparing classifiers under repeated resampling we follow Nadeau and Bengio [55].

### III. DEPLOYMENT CRITERIA AND THREAT MODEL

This section defines the system and threat model and the three quantities the evaluation measures, then combines them into a deployability test.

#### A. SYSTEM AND THREAT MODEL

A near-real-time RIC serves a set of E2 nodes (Fig. 1), with user devices, IoT devices among them, attached through those nodes. The adversary controls a subset of devices and uses them for flooding, scanning, brute-force login, or web attacks, the categories present in our corpora. The adversary does not control the RIC, the E2 interface, or the xApp. The defender's xApp consumes either radio telemetry windows delivered over E2SM-KPM or flow records produced by a flow exporter co-located with the user plane, and may act through E2SM-RC. Because E2 does not carry packets, a flow-level detector needs such an exporter outside the RIC; its cost and placement are discussed in Section VII-A.

#### B. CRITERION 1: GENERALIZATION ACROSS DEPLOYMENTS

Let  $\mathcal{D}_A$  and  $\mathcal{D}_B$  be independently collected corpora over a shared feature space. A model  $f_\theta$  trained on  $\mathcal{D}_A$  is characterized by the gap between its held-out score on  $\mathcal{D}_A$  and its score on  $\mathcal{D}_B$ . Because the two corpora have different attack prevalence (Table 2), a gap in macro- $F_1$  mixes a change in the classifier's behavior with a change in class balance: even a classifier that ignores its input changes macro- $F_1$  when prevalence changes. We therefore report the gap in balanced accuracy, which a change of prevalence cannot move,

$$\Delta_{BA} = \text{BA}(f_\theta; \mathcal{D}_A^{\text{test}}) - \text{BA}(f_\theta; \mathcal{D}_B), \quad (1)$$

together with the macro- $F_1$  gap  $\Delta_{F_1}$  and its difference from the gap of the majority-class baseline. We quantify covariate shift feature by feature as

$$\bar{W}_1 = \frac{1}{|\mathcal{F}|} \sum_{j \in \mathcal{F}} w_1 \left( Q_j(P_A^{(j)}), Q_j(P_B^{(j)}) \right), \quad (2)$$

where  $Q_j$  is the quantile transform of feature  $j$  fitted on source training data only. Normalizing each corpus by its own quantiles would map both marginals to the uniform distribution and make every term zero.

#### C. CRITERION 2: ALERT BURDEN UNDER A REALISTIC BASE RATE

Let  $\lambda_b$  be the benign arrival rate in the detector's own unit: flow records per hour for a flow-level detector, and telemetry windows per hour for a window-level detector. A benign UE produces 225 windows of 16 s per hour. The false alerts per hour are

$$A(\tau) = \lambda_b \text{FPR}(\tau), \quad (3)$$

and, for an operational attack prevalence  $\pi$ , the fraction of alerts that are true is

$$\text{PPV}(\tau) = \frac{\pi R(\tau)}{\pi R(\tau) + (1 - \pi) \text{FPR}(\tau)}. \quad (4)$$

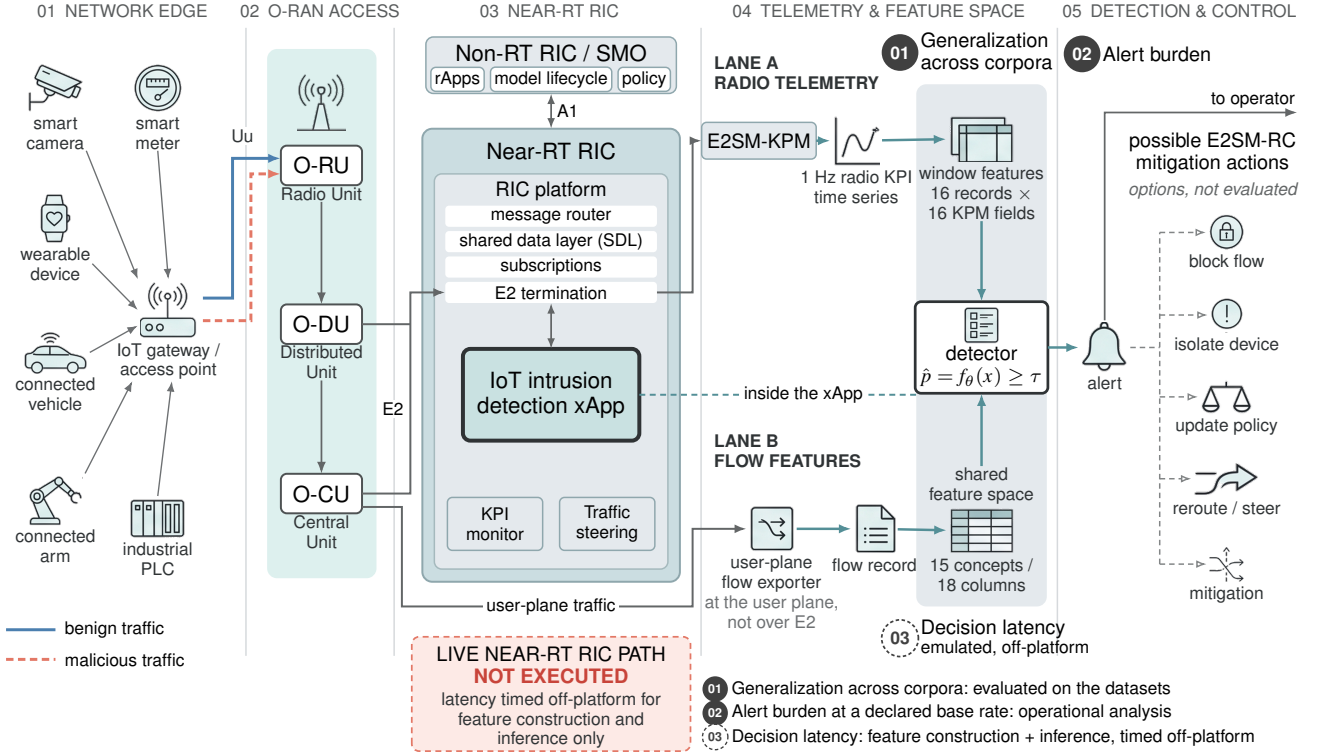
Neither  $\pi$  nor  $\lambda_b$  can be measured from the corpora; both are declared and swept. PPV depends on  $\pi$  only, and  $A$  scales linearly in  $\lambda_b$ , so we also report false alerts per  $10^6$  benign flows, which does not depend on  $\lambda_b$ .

#### D. CRITERION 3: LATENCY AGAINST THE CONTROL BUDGET

The decision latency of an xApp decomposes as

$$L = t_{\text{ind}} + t_{\text{feat}} + t_{\text{inf}} + t_{\text{act}} + t_q, \quad (5)$$

for indication decoding, feature construction, inference, control encoding, and queueing. We measure  $t_{\text{feat}}$  and  $t_{\text{inf}}$  only; the other three terms need a RIC in the path. Conformance is a tail property: for budget  $B$  and tolerated violation rate  $\varepsilon$  we require  $q_{1-\varepsilon}(L) \leq B$ . The O-RAN E2 specification sets near-real-time latency requirements anywhere from 10 ms to 1 s [17], so we report the smallest conforming budget rather than a verdict at one chosen value.



**FIGURE 1.** Reference O-RAN edge deployment and the detection loop. All device traffic reaches the radio unit over Uu. The detection xApp receives radio telemetry through the E2 termination (lane A) and flow records from an exporter at the user plane, mapped onto the shared feature space (lane B); column 04 expands the xApp's own feature construction and inference, with one detector per layer. Numbered markers show what this paper measures: (01) generalization across corpora and (02) alert burden are evaluated on the corpora; (03) decision latency is timed off-platform on one host, with no RIC in the path. The mitigation actions on the right are options E2SM-RC offers and are not evaluated.

### E. A DEPLOYABILITY TEST

The three criteria combine into a test that a detector passes or fails:

$$\begin{aligned} \mathcal{P}(f_\theta) = & \mathbf{1}[\bar{\Delta}_{BA} \leq \delta] \cdot \mathbf{1}[\bar{q}_{0.99}(L) \leq B] \\ & \cdot \mathbf{1}[\exists \tau : R(\tau) \geq r_{\min}, A(\tau) \leq A_{\max}, \\ & \text{PPV}(\tau) \geq \rho], \end{aligned} \quad (6)$$

where  $\bar{\Delta}_{BA}$  and  $\bar{q}_{0.99}$  are upper 95% confidence bounds, so a pass cannot rest on a lucky point estimate, and every alert condition is evaluated on the target corpus. The minimum recall  $r_{\min}$  prevents the alert term from being satisfied by a detector that never alerts. The operating threshold is the most sensitive admissible one,

$$\begin{aligned} \tau^* = \arg \max_{\tau} R(\tau) \quad \text{s.t.} \quad & R(\tau) \geq r_{\min}, A(\tau) \leq A_{\max}, \\ & \text{PPV}(\tau) \geq \rho. \end{aligned} \quad (7)$$

We propose this test as a reporting device, not as an O-RAN requirement: its bounds are operator requirements, and Section VII-B states the values we use and how the verdict depends on them.

### IV. METHODOLOGY

This section describes the corpora and feature spaces, the detectors, the evaluation protocols, and the statistical analysis.

### A. CORPORA

$\mathcal{D}_A$  is NetsLab-5GORAN-IDD [39], captured on a physical OpenAirInterface O-RAN testbed with benign traffic and five attack categories: denial of service (DoS), distributed denial of service (DDoS), probing, brute force, and web attacks. It provides two layers that cannot be joined record by record from the published files: flow records exported by Zeek from the central unit, and 1 Hz radio telemetry from the distributed unit. We use both. The radio layer is aggregated into windows of 16 consecutive records of one UE, summarized by the mean and standard deviation of 16 KPM fields. Its 30 capture sessions are recovered from gaps of more than five minutes in the clock. Fard et al. describe the corpus as 42 experiment runs [31]; a gap rule of 90 to 105 s recovers exactly 42 segments, each with a single attack category, and each of our sessions is a union of whole segments. The five-minute rule merges runs recorded back to back (10 of the 30 sessions hold two or three attack subcategories), which keeps such runs on the same side of every split. Each recovered session carries a single category label in every record, so a radio label is a session label: every window of an attack session is an attack window. The sessions follow the capture schedule (Section VI-B): nine benign sessions on days 0 to 6, the attack categories in blocks on days 47 to 53, and one benign session at the end.  $\mathcal{D}_B$  is 5G-NIDD [37], [38], captured on a 5G test network at a different institution, with flow records exported

**TABLE 2. Datasets.**  $\mathcal{D}_A$  supplies training and in-distribution testing on both layers.  $\mathcal{D}_B$  and  $\mathcal{D}_C$  are transfer targets; each also trains its own in-target references, and  $\mathcal{D}_B$  trains the reverse direction.

|                   | $\mathcal{D}_A$ network | $\mathcal{D}_A$ radio | $\mathcal{D}_B$ | $\mathcal{D}_C$ |
|-------------------|-------------------------|-----------------------|-----------------|-----------------|
| Unit              | flow                    | 16 s UE window        | flow            | flow            |
| Source            | Zeek                    | KPM, 1 Hz             | Argus           | NFStream        |
| Samples           | 1 640 182               | 2 808                 | 1 215 889       | 39 425          |
| Features          | 18                      | 32                    | 18              | 18              |
| Attack prevalence | 94.63%                  | 76.46%                | 60.71%          | 2.13%           |
| Attack categories | 5                       | 5                     | 2               | 3               |
| Groups            | 318 hosts               | 30 runs               | 20 files        | 3 files         |
| Role              | train, test             | train, test           | target          | target          |

by Argus and two attack categories, DoS and probing. Its published files carry no addresses, ports, or timestamps, but they keep Argus’s record counter and file offset. The offset falls back to its starting value at every file boundary, which recovers 20 capture files: two passes over the same ten captures, one per base station (the published `Combined.csv` also restarts its row index where the second pass begins). Every attack file mixes benign traffic with one attack type. A third corpus,  $\mathcal{D}_C$ , is a second transfer target from a third site: the flow records of a 5G core testbed (Open5GS in Docker) published by Nugraha et al. [22]. Only three of its files are flow records, each with one attack (a SYN flood, an ICMP flood, and a PFCP session-deletion attack), 39,425 flows with 839 attacks; its benign flows are user traffic and the core’s own signalling. It was exported by NFStream, a third flow exporter, which counts link-layer bytes and, for flows inside the GTP-U user-plane tunnel, the 44 bytes of encapsulation; we remove both per packet so that the shared columns hold the inner packet’s IP bytes, as for  $\mathcal{D}_A$ . Table 2 summarizes the datasets.

## B. SHARED FEATURE SPACE

Zeek and Argus share no column names, so the flow feature space is defined by meaning (Table 3): 15 concepts realized as 18 columns, namely seven counts each exporter reports, a one-hot protocol with a fixed level list, and seven derived quantities computed identically on both sides. One mapping deserves stating because the obvious choice is wrong: Zeek’s `src_bytes` counts application payload, which is zero for most floods and scans, whereas Argus’s `SrcBytes` counts whole packets. The counterpart of `SrcBytes` is Zeek’s `src_ip_bytes`, at 40 against 42 bytes per packet. A floor on duration and  $\log(1 + x)$  on unbounded quantities are applied identically to both corpora; scaling is fitted on source training data only. No address, port, or timestamp is used as a feature.

We also define, before any result under it was computed, an exporter-robust subset of nine columns: the protocol and the five per-packet means and direction ratios. The two exporters disagree about where a flow ends, which moves duration, rates, and totals. Per-packet means and direction ratios move less, but they are not invariant: cutting a flow changes them whenever its packets differ, for example when handshake

**TABLE 3. Shared flow feature space, generated from `configs/features/shared_space.yaml`. The exporter-robust subset is the protocol one-hot, the three mean packet sizes, and the two direction ratios.**

| Concept           | $\mathcal{D}_A$ (Zeek)                      | $\mathcal{D}_B$ (Argus) | Role        |
|-------------------|---------------------------------------------|-------------------------|-------------|
| duration          | duration                                    | Dur                     | base        |
| src_bytes         | src_ip_bytes                                | SrcBytes                | base        |
| dst_bytes         | dst_ip_bytes                                | DstBytes                | base        |
| src_pkts          | src_pkts                                    | SrcPkts                 | base        |
| dst_pkts          | dst_pkts                                    | DstPkts                 | base        |
| tot_pkts          | src_pkts + dst_pkts                         | TotPkts                 | base        |
| tot_bytes         | src_ip_bytes + dst_ip_bytes                 | TotBytes                | base        |
| proto             | proto                                       | Proto                   | one-hot (4) |
| mean_pkt_size     | tot_bytes / max(tot_pkts, 1)                |                         | derived     |
| src_mean_pkt_size | src_bytes / max(src_pkts, 1)                |                         | derived     |
| dst_mean_pkt_size | dst_bytes / max(dst_pkts, 1)                |                         | derived     |
| bytes_per_s       | tot_bytes / max(duration, duration_floor_s) |                         | derived     |
| pkts_per_s        | tot_pkts / max(duration, duration_floor_s)  |                         | derived     |
| src_byte_ratio    | src_bytes / max(tot_bytes, 1)               |                         | derived     |
| src_pkt_ratio     | src_pkts / max(tot_pkts, 1)                 |                         | derived     |

and teardown packets fall into different pieces, and the two exporters count the same packet at 40 and 42 bytes, a fixed offset of about 5% for minimum-size packets.

## C. DETECTORS

We evaluate six supervised architectures and two trivial baselines on every protocol: LR, a depth-12 DT, an RF [56], XGB [57], scikit-learn’s HGB [58], and an MLP, against a majority-class predictor and a stratified sampler of the training prior. The baselines ignore their input, so any change in their score between two protocols is a change in class composition alone. Every supervised model receives the same class-imbalance policy: balanced class weights, applied as sample weights for the MLP and as `scale_pos_weight` for XGB. Hyperparameters are fixed, not tuned (Table 4): tuning on held-out source data would reward structure specific to the source, which is the effect the transfer experiments measure, and tuning on the target would leak it.

Three further families answer specific questions. Two sequence models, a GRU [59] and a one-dimensional convolutional network, read the raw 16-step KPM sequences of the same radio windows. Two unsupervised detectors, an isolation forest [60] and an autoencoder, are trained on benign source flows only, with the alarm threshold fixed at the 99th percentile of source-benign scores. A CORAL arm [54] recolors source flows to the covariance of the unlabeled target before training; it is the only experiment in which a detector sees target data.

## D. EVALUATION PROTOCOLS

Splits are group-disjoint wherever a group key exists, and a guard raises if a group appears on both sides. On the radio layer the key is the capture session, and we compare three protocols: a random split of windows; a run-disjoint split; and a category-stratified run-disjoint split that keeps sessions of every attack category on both sides, so that no test category is unseen in training. Two further designs address time (Section VI-B): the latest or the earliest session of every category

**TABLE 4.** Hyperparameters, identical across protocols and seeds.

| Model            | Settings                                                                             |
|------------------|--------------------------------------------------------------------------------------|
| Majority, Strat. | training majority; samples of the training prior                                     |
| LR               | max_iter 2000, balanced weights, standardized                                        |
| DT               | depth 12, balanced weights                                                           |
| RF               | 200 trees, min. leaf 2, balanced subsample weights                                   |
| XGB              | 300 trees, depth 6, $\eta = 0.1$ , subsample 0.8, hist                               |
| HGB              | 300 iterations, depth 6, $\eta = 0.1$ , balanced weights                             |
| MLP              | 128–64, Adam, early stopping (patience 10),<br>balanced sample weights, standardized |
| GRU              | hidden 32; BCE with positive weight $n_-/n_+$                                        |
| 1D-CNN           | two convolutions (32, kernel 3), average pooling                                     |
| IF               | 200 trees; benign-only training                                                      |
| AE               | 12–6–12 MLP, reconstruction error; benign-only                                       |
| CORAL            | per-domain z-score, identity-regularized covariances                                 |

held out, against draws that hold out a random session of every category, and each benign session held out alone. On the flow layer the key is the source address (318 groups); each of 20 split seeds draws a 300,000-flow subsample of  $\mathcal{D}_A$  and a source-address-disjoint split of it. Every model is then evaluated on the whole of  $\mathcal{D}_B$ , and the reverse direction trains on a random split of  $\mathcal{D}_B$  and evaluates on  $\mathcal{D}_A$ . On  $\mathcal{D}_B$  itself, in-target references hold out four of the 20 capture files, or train on one base station’s pass and test on the other’s (Section VI-C). Except for these references, CORAL, the domain classifier, and the prior estimates of Section VI-H, all of which are labeled as such, no transform, threshold, or model choice uses  $\mathcal{D}_B$ , and every read of it is logged.

### E. STATISTICAL ANALYSIS

Uncertainty is estimated over 20 split seeds. The splits resample one population of 30 sessions or 318 source addresses, so their training and test sets overlap and a paired  $t$ -test over split means understates the variance. We therefore report Nadeau–Bengio corrected intervals [55], which scale the variance by  $1/J + n_{\text{test}}/n_{\text{train}}$  for  $J$  splits. Holm correction is applied within each family of tests, where a family is one metric and one comparison (a protocol pair or a transfer direction) across the six supervised architectures; the two sequence models form a family of their own. Operating points are computed from confusion counts pooled over folds; Section VI-G shows why averaging per-fold precision is wrong. Pooled counts repeat a sample in every seed that tests it, and samples of one cluster are correlated, so a binomial interval on them is far too narrow. Intervals for pooled operating points come instead from a cluster bootstrap with 2,000 replicates that resamples, with replacement, both the split seeds and the clusters: radio sessions,  $\mathcal{D}_A$  source addresses, or  $\mathcal{D}_B$  capture files.

### F. EXPERIMENTAL PIPELINE AND INFERENCE LOOP

Fig. 2 summarizes the pipeline and Algorithm 1 gives the decision loop an xApp would run. The algorithm describes the intended control logic; we did not deploy it in a RIC, and the latency measurements of Section VI-I time its feature construction and inference stages only. A decision that misses the budget is still acted on and recorded as a violation, since

### Algorithm 1 Budget-aware decision loop of a detection xApp

**Require:** indication stream  $\mathcal{S}$ , model  $f_\theta$ , threshold  $\tau$ , budget  $B$ , margin  $\delta_m$

**Ensure:** E2 control actions; violation count  $v$ ; latency record

```

1:  $\mathcal{W} \leftarrow$  empty per-UE window map;  $v \leftarrow 0$ 
2: for all indication  $m$  arriving on  $\mathcal{S}$  do
3:    $t_0 \leftarrow \text{Now}()$ ;  $u \leftarrow \text{UEOf}(m)$ 
4:    $\mathcal{W}[u] \leftarrow \text{UpdateWindow}(\mathcal{W}[u], m)$ 
5:    $x \leftarrow \text{Features}(\mathcal{W}[u])$ 
6:    $\hat{p} \leftarrow f_\theta(x)$ 
7:   if  $\hat{p} \geq \tau$  then
8:     SendE2(EncodeControl( $u, \hat{p}$ ))  $\triangleright$  even if late
9:   end if
10:   $\ell \leftarrow \text{Now}() - t_0$ 
11:  if  $\ell > B - \delta_m$  then
12:     $v \leftarrow v + 1$   $\triangleright$  deadline violation, recorded
13:  end if
14:  RecordLatency( $\ell$ )  $\triangleright$  every indication
15: end for
```

a late mitigation is usually better than none; an operator who prefers to drop late actions moves the send inside the deadline check.

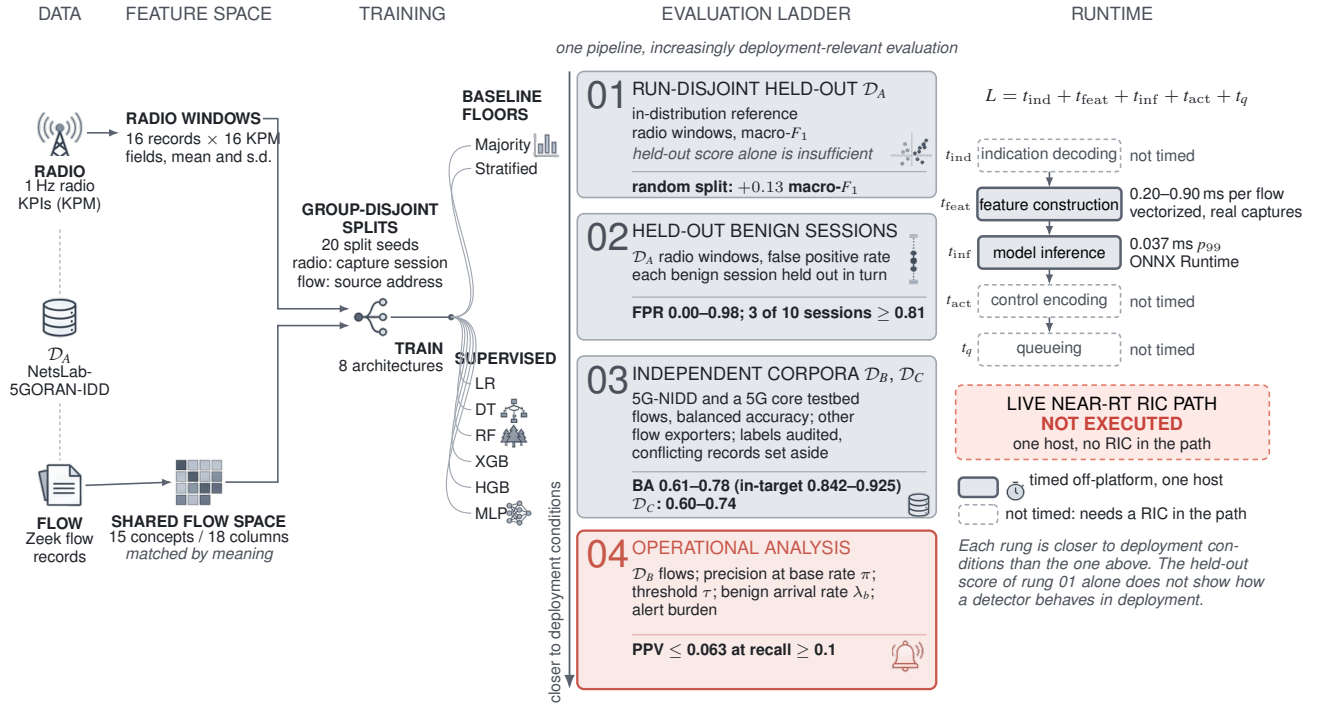
### V. EXPERIMENTAL SETUP

All measurements were taken on one workstation: a 13th Gen Intel Core i9-13900H (14 cores, 20 threads), 47.6 GB of memory, Windows 11 Pro build 26120, Python 3.11.9, scikit-learn 1.8.0, XGBoost 3.2.0, NumPy 2.3.5 and ONNX Runtime 1.30.0; the sequence models used PyTorch 2.14 on the CPU in a separate environment. A complete package list is committed with the results. No RIC was involved: we did not deploy the detector as an xApp, run an E2 agent, or containerize anything, and all latency figures are emulated.

The radio-layer protocol comparison uses 20 split seeds with two model seeds each; the other radio-layer and flow-layer experiments use 20 split seeds with one model seed; calibration uses 10 split seeds of a three-way source split into training, calibration, and test groups. The declared operational parameters are an attack prevalence  $\pi = 0.002$  and a benign arrival rate  $\lambda_b = 240,000$  flows per hour, both swept; for the radio layer we report false alerts per benign UE-hour. Neither is taken from an operator. The rate  $\lambda_b$ , about 67 flows per second, is below the several hundred flows per second that Jin et al. expect at a heavily loaded base station of a large LTE network [61], so the alert volumes we report are low rather than high for an edge site that serves several cells. Latency is timed per call with a monotonic clock, with garbage collection enabled, after 1,000 warm-up calls, over 20,000 calls, on one thread per library; distribution-free confidence intervals are given for every quantile.

### VI. RESULTS

The results follow the evaluation ladder: protocol sensitivity within one corpus, time order and benign sessions, the target



**FIGURE 2.** Experimental pipeline and evaluation ladder. Radio windows and flows from  $D_A$  are summarized separately, split by group, and used to train eight architectures. Each rung of the ladder moves closer to deployment conditions: a run-disjoint held-out split, held-out benign sessions, two independently collected corpora evaluated on flows with consistent labels, and an operational analysis at a declared base rate. The bold line under each rung gives the measured result. In the runtime column only feature construction and inference were timed, on one host with no RIC in the path; the live-RIC arm was not executed.

corpus and a published result on it, transfer across corpora and what the gap is made of, a third corpus, then alert burden, calibration, and latency.

### A. EVALUATION-PROTOCOL SENSITIVITY

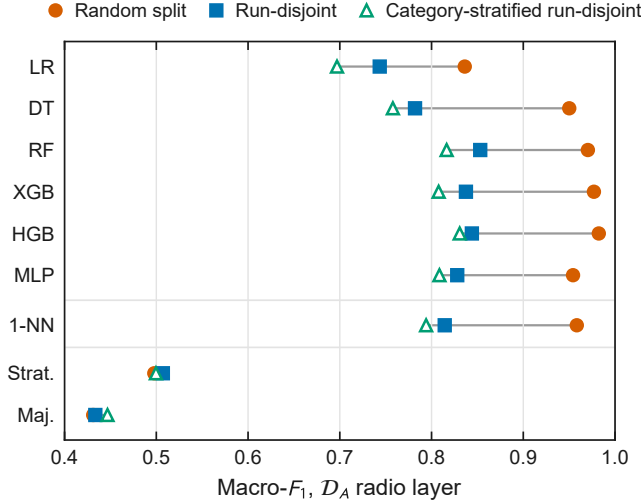
Table 5 and Fig. 3 compare three protocols on the radio layer. A random split raises macro- $F_1$  over a run-disjoint split by 0.09 to 0.17 across the six architectures, 0.13 on average, and the gain is positive in at least 16 of 20 splits for every architecture. The trivial baselines gain nothing. The corrected interval for the average gain is  $[-0.01, 0.27]$  ( $p = 0.06$ ): with 30 capture sessions the size of the effect cannot be pinned down, and no single architecture's gain survives Holm correction. A naive paired  $t$ -test would report  $p = 6 \times 10^{-5}$  for the same data.

The gain does not come from test categories absent in training: in all 20 run-disjoint splits every test category also appeared in training. The category-stratified protocol guarantees this by construction and gives a gain of 0.14 to 0.19; it moves sessions of rare categories into the test set, which makes its test sets harder, so we do not read its larger gain as further evidence. A model-free probe points to the mechanism. Under a random split, 68% of test windows have their nearest training window in the same capture session (65% to 71% across splits), and a one-nearest-neighbor (1-NN) lookup scores 0.96 macro- $F_1$  against 0.81 under a run-disjoint split (Table 5). Because every radio label is a session label, recognizing the session is enough to predict the label, and

**TABLE 5.** Protocol sensitivity on the  $D_A$  radio layer: macro- $F_1$  under random, run-disjoint (RD) and category-stratified run-disjoint (CS) splits, and paired gains of the random split with Nadeau-Bengio corrected 95% intervals over 20 split seeds. No supervised architecture's gain is significant after Holm correction. 1-NN: a model-free nearest-neighbor lookup on the same splits.

| Model  | Random | Run-disj. | Cat.-strat. | $\Delta_{R-RD}$ [95% CI] | $\Delta_{R-CS}$ [95% CI] |
|--------|--------|-----------|-------------|--------------------------|--------------------------|
| LR     | 0.836  | 0.744     | 0.697       | +0.093 [-0.05, +0.24]    | +0.139 [-0.06, +0.33]    |
| DT     | 0.950  | 0.782     | 0.758       | +0.168 [+0.02, +0.32]    | +0.192 [+0.02, +0.36]    |
| RF     | 0.970  | 0.853     | 0.816       | +0.117 [-0.04, +0.27]    | +0.154 [-0.02, +0.32]    |
| XGB    | 0.977  | 0.837     | 0.808       | +0.140 [-0.00, +0.28]    | +0.169 [-0.02, +0.35]    |
| HGB    | 0.982  | 0.844     | 0.831       | +0.138 [-0.01, +0.28]    | +0.152 [-0.03, +0.33]    |
| MLP    | 0.954  | 0.828     | 0.809       | +0.126 [-0.01, +0.26]    | +0.146 [-0.02, +0.31]    |
| Strat. | 0.498  | 0.507     | 0.500       | -0.009 [-0.01, -0.00]    | -0.002 [-0.01, +0.01]    |
| Maj.   | 0.431  | 0.434     | 0.447       | -0.002 [-0.02, +0.01]    | -0.016 [-0.04, +0.01]    |
| 1-NN   | 0.958  | 0.814     | 0.794       | +0.144 [+0.00, +0.28]    | +0.164 [+0.01, +0.31]    |

a random split lets any model, including a lookup table, do so. Under the run-disjoint split the lookup scores above LR and DT and within 0.04 of the best supervised architecture, so on this layer the learned models add little to the retrieval of similar windows. The two sequence models of Table 6, which read the raw 16-step KPM sequences instead of window summaries, show the same pattern more clearly: a random split scores 0.923 to 0.944 against 0.755 to 0.759 under a run-disjoint split, a gain of 0.16 to 0.19 whose corrected intervals exclude zero for both models (Holm-adjusted  $p \leq 0.02$  within the pair).



**FIGURE 3.** Macro- $F_1$  on the  $\mathcal{D}_A$  radio layer under three protocols, mean over 20 split seeds; the gray line spans the three protocols of each model. The random split raises every supervised model and the model-free 1-NN lookup alike, and the input-blind baselines do not move. Keeping every attack category in training (category-stratified) does not remove the drop. Corrected intervals are in Table 5.

**TABLE 6.** Sequence models on the same radio windows and splits as Table 5.

| Model  | Random | Run-disj. | Cat.-strat. | $\Delta_{R-RD}$ [95% CI] |
|--------|--------|-----------|-------------|--------------------------|
| 1D-CNN | 0.944  | 0.755     | 0.763       | +0.189 [+0.05, +0.32]    |
| GRU    | 0.923  | 0.759     | 0.769       | +0.165 [+0.04, +0.29]    |

## B. TIME ORDER AND BENIGN SESSIONS

The radio layer spans 52.96 days, but its capture order follows the scenario schedule (Table 7). Nine benign sessions were recorded on days 0 to 6, the five attack categories in blocks on days 47 to 53, and one last benign session on day 53. Every radio label is a session label: a session is benign, or one attack category, throughout. A split on session order thus holds out whole attack categories. At training fractions of 0.4 to 0.7, the latest sessions contain 1 to 4 attack categories absent from training, and a single benign session. Such a split measures the detection of unseen categories, so we do not report it as drift.

To hold category coverage fixed, we held out the latest session of every category, benign included, and compared the result with 50 draws that hold out a random session of every category (Table 8). With the latest sessions held out, balanced accuracy is 0.52 to 0.57, against control means of 0.75 to 0.83, yet 12% to 32% of control draws score as low. With the earliest sessions held out, it is 0.95 to 0.99, which 4% to 34% of control draws match or exceed. Time order does not move the result outside ordinary session-to-session variation.

The benign session in the test set decides it. Held out one at a time (Table 7 and Fig. 4), 6 of the ten benign sessions draw a false positive rate of at most 0.07, while sessions 4, 8 and 29 (days 3, 6 and 53) draw 0.81 to 0.98. The last benign session, the only one recorded during the attack period, draws 0.87, or

**TABLE 7.** False positive rate on each benign radio session when that session alone is held out and the models train on all other 29 sessions, mean and range over the six architectures. Day: capture start relative to the first session. Alerts per benign UE-hour:  $FPR \times 225$  windows.

| Session | Day  | UEs | Windows | FPR  | Range        | Alerts/UE-h |
|---------|------|-----|---------|------|--------------|-------------|
| 0       | 0.0  | 1   | 57      | 0.00 | [0.00, 0.00] | 0           |
| 1       | 0.0  | 1   | 54      | 0.02 | [0.00, 0.06] | 6           |
| 2       | 2.8  | 1   | 58      | 0.00 | [0.00, 0.00] | 0           |
| 3       | 2.9  | 1   | 55      | 0.07 | [0.02, 0.20] | 15          |
| 4       | 3.0  | 1   | 24      | 0.81 | [0.54, 0.96] | 181         |
| 5       | 3.9  | 1   | 56      | 0.14 | [0.00, 0.77] | 31          |
| 6       | 4.0  | 2   | 60      | 0.00 | [0.00, 0.00] | 0           |
| 7       | 4.0  | 1   | 95      | 0.01 | [0.00, 0.03] | 2           |
| 8       | 6.0  | 1   | 56      | 0.98 | [0.93, 1.00] | 220         |
| 29      | 52.9 | 3   | 146     | 0.87 | [0.84, 0.93] | 196         |

**TABLE 8.** Category-stratified time split on the  $\mathcal{D}_A$  radio layer: balanced accuracy when the latest or the earliest session of every category is held out, against the mean of 50 draws that hold out a random session of every category. "% ctrl.  $\leq$ ": share of control draws scoring at or below the time split. Every test category is in training.

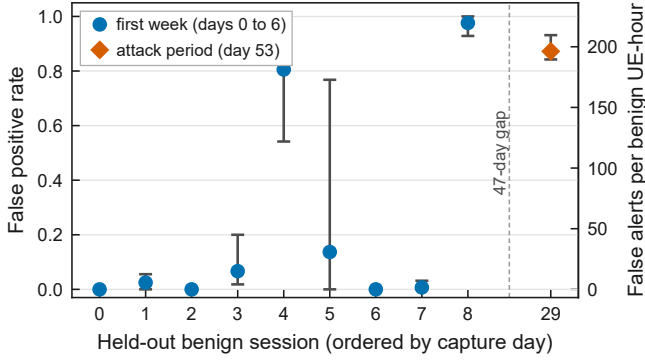
| Model | Control |                | Latest held out |                | Earliest held out |                |
|-------|---------|----------------|-----------------|----------------|-------------------|----------------|
|       | BA      | % ctrl. $\leq$ | BA              | % ctrl. $\leq$ | BA                | % ctrl. $\leq$ |
| LR    | 0.749   | 0.573          | 0.20            | 0.950          | 0.96              |                |
| DT    | 0.774   | 0.521          | 0.12            | 0.962          | 0.84              |                |
| RF    | 0.811   | 0.562          | 0.32            | 0.970          | 0.66              |                |
| XGB   | 0.812   | 0.567          | 0.26            | 0.978          | 0.72              |                |
| HGB   | 0.817   | 0.550          | 0.32            | 0.995          | 0.94              |                |
| MLP   | 0.827   | 0.574          | 0.18            | 0.983          | 0.94              |                |

196 false alerts per benign UE-hour, but two sessions from the first week draw as much. With ten benign sessions, a change over time cannot be separated from this heterogeneity. The practical consequence is that a detector's false positive rate on a new site cannot be read off its rate on the benign sessions it was tested on.

## C. THE TARGET CORPUS: LABELS AND A PUBLISHED RESULT

**Label conflicts.** Before transferring to  $\mathcal{D}_B$ , we asked whether its labels are a function of its flow records. For half of the corpus they are not. Of the 1,215,889 flows, 52% (632,378) share every field except the row index, Seq, and Offset with a flow of the opposite label (Table 9, Fig. 5). All of them lie in the two UDP-flood captures, 75% of file 5 and 98% of file 15, one from each base station. The pattern is specific: each of the 281,529 flows labeled UDP flood in file 15 has an identical copy labeled benign in file 5, in equal numbers for 33,704 of the 33,708 distinct records. The same traffic appears to have been captured at both stations and labeled differently at each; the copies make up 59% of the corpus's benign class. We cannot tell from the published files which label is correct, so every  $\mathcal{D}_B$  result below is given with and without these benign copies, and no label is changed.

The conflicts set a ceiling on every classifier that reads the record. The best any function of the 90 content fields can do, even evaluated on the flows it was built from, is a balanced accuracy of 0.766; in the 18 shared columns the ceiling is 0.753, which is where DT, XGB, HGB, and the MLP sit when trained and tested on a random split of  $\mathcal{D}_B$  (Table 10). With



**FIGURE 4.** False positive rate on each benign radio session when that session alone is held out: mean (marker) and range (bar) over the six architectures, sessions in capture order. The right axis gives the same rate as false alerts per benign UE-hour (225 windows). Sessions of the same week range from no false positive to almost all windows flagged, so the session a detector meets, rather than the time of capture, sets its false positive rate.

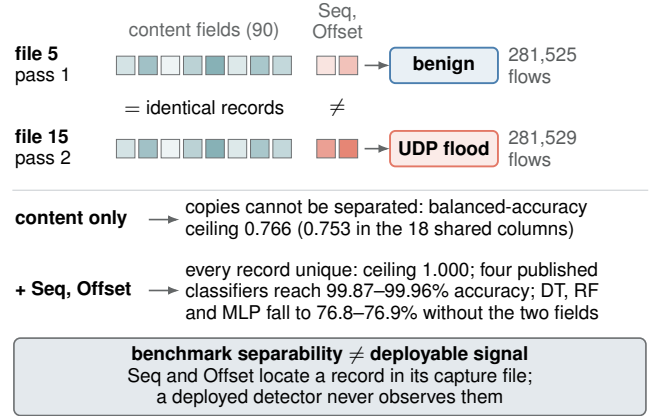
**TABLE 9.** Label conflicts in  $\mathcal{D}_B$  (5G-NIDD). Content: all fields of the published encoding except row index, Seq, Offset, and the labels. Ceiling: balanced accuracy of the best lookup on distinct records, in sample.

| Features              | Cols | Records   | % in conflicting records |        |        | BA ceiling |
|-----------------------|------|-----------|--------------------------|--------|--------|------------|
|                       |      |           | All                      | Benign | Attack |            |
| Content fields        | 90   | 374 946   | 52.0                     | 58.9   | 47.5   | 0.766      |
| Shared space          | 18   | 246 281   | 62.4                     | 59.7   | 64.1   | 0.753      |
| Content + Seq, Offset | 92   | 1 215 869 | 0.0                      | 0.0    | 0.0    | 1.000      |

Seq and Offset added, every record becomes unique and the ceiling is 1.000.

*References inside the target.* To separate what transfer costs from what  $\mathcal{D}_B$  itself allows, we trained and tested the six architectures on  $\mathcal{D}_B$  alone (Table 10). On a random split they reach 0.708 to 0.752 in the shared space, the ceiling above, and 0.842 to 0.925 when the benign copies are left out of the test set. On held-out capture files they reach 0.931 to 0.982 in the shared space and 0.990 to 1.000 with native features; none of the three draws held out a UDP-flood capture. Across the two base stations, within one corpus and one exporter, balanced accuracy falls to 0.618 to 0.641 from station 1 to station 2 and 0.639 to 0.650 in the other direction. Both failures come from the copies: trained on station 1, whose labels call the flood copies benign, a model misses station 2's floods; trained on station 2, it flags station 1's benign-labeled copies, and without them in the test set it scores 0.973 to 0.988. On flows whose label the record determines, then, the shared space separates  $\mathcal{D}_B$  almost perfectly, and the target is not what limits transfer.

*A published result down the ladder.* The dataset's authors report binary accuracy of 99.85% to 99.95% on a random 70/30 split [11]. We reproduced their pipeline: the published one-hot encoding, the ten features with the highest ANOVA  $F$ -score,  $z$ -scoring, and their five classifiers with scikit-learn defaults, since the tuned values are not reported. Our  $F$ -scores rank the features in exactly their order, and on their split we obtain 99.87% to 99.96% accuracy for DT, RF,



**FIGURE 5.** The label conflict in  $\mathcal{D}_B$ . The same flow records occur in two capture files of the two passes, labeled benign in one and UDP flood in the other; the tiles are schematic. No function of the record's content can separate the copies, which caps balanced accuracy. The record-position fields Seq and Offset differ between the copies, make every record unique, and carry the published 99.9% result; a deployed detector never observes them.

KNN, and the MLP (Naive Bayes 92.71%), matching their results (Table 11). Their first and second features are Seq and Offset, a record's position in its capture file, which no deployed detector can observe and which, by the ceilings above, is what separates the conflicting records. Replacing only these two with the next two features of their ranking drops DT, RF, and the MLP to 76.8% to 76.9% accuracy (balanced accuracy 0.704 to 0.706, false positive rate 0.588 to 0.591). The drop is the label conflict, not a loss of attack signal: on the test flows without the benign copies, RF still scores 0.9999. The two counters let the published models tell identical records apart by where they sit in their file. Held out by capture file, the published features average 0.859 to 0.923 for DT, RF, and the MLP, with RF ranging from 0.82 to 1.00 across folds; without the counters the average is 0.799 to 0.800 and RF ranges from 0.50 to 1.00, with accuracy as low as 1.6% on the fold that tests one flood capture after training on the other. Across base stations the published features give 0.762 to 0.895, and the features without counters 0.656 to 0.692. A published 99.9% on this corpus says little about a new capture file or a new base station, and nothing about a deployment, where record positions do not exist.

#### D. TRANSFER ACROSS CORPORA

Table 12 and Fig. 6 report transfer from  $\mathcal{D}_A$  to  $\mathcal{D}_B$ . In macro- $F_1$  every supervised model loses 0.10 to 0.13, but the majority baseline, which reads no features, loses 0.11 as well, because  $\mathcal{D}_B$  has a lower attack prevalence than  $\mathcal{D}_A$ . Measured against that baseline the macro- $F_1$  loss is between -0.018 and +0.015, and after correction none of the macro- $F_1$  losses is significant. Macro- $F_1$  is the wrong instrument here. Balanced accuracy, which prevalence cannot move, shows the loss clearly: from 0.81–0.91 on held-out source data to 0.53–0.63 on  $\mathcal{D}_B$ , a drop of 0.21 to 0.36 that is significant for 5 of six architectures after correction, is positive in at least 18 of 20 splits for every

**TABLE 10.** References inside  $\mathcal{D}_B$ : balanced accuracy of models trained and tested on  $\mathcal{D}_B$  only, mean over  $n$  draws of a 240,000-flow training sample. \*Range over the six architectures on the test flows without the benign copies of Table 9. Capture-file draws hold out four of the 20 files.

| Features    | Split          | LR    | DT    | RF    | XGB   | HGB   | MLP   | BA*       | $n$ |
|-------------|----------------|-------|-------|-------|-------|-------|-------|-----------|-----|
| Shared (18) | Random         | 0.708 | 0.752 | 0.723 | 0.752 | 0.752 | 0.750 | 0.84–0.92 | 3   |
| Shared (18) | Capture file   | 0.955 | 0.982 | 0.940 | 0.943 | 0.956 | 0.931 | 0.93–0.98 | 3   |
| Shared (18) | Station 1 to 2 | 0.618 | 0.638 | 0.640 | 0.641 | 0.640 | 0.635 | 0.62–0.64 | 1   |
| Shared (18) | Station 2 to 1 | 0.639 | 0.648 | 0.648 | 0.648 | 0.648 | 0.650 | 0.97–0.99 | 1   |
| Native      | Random         | 0.710 | 0.764 | 0.735 | 0.764 | 0.764 | 0.764 | 0.80–0.87 | 2   |
| Native      | Capture file   | 0.990 | 1.000 | 0.999 | 1.000 | 1.000 | 0.994 | 0.99–1.00 | 2   |
| Native      | Station 1 to 2 | 0.698 | 0.661 | 0.662 | 0.662 | 0.662 | 0.662 | 0.66–0.70 | 1   |
| Native      | Station 2 to 1 | 0.648 | 0.648 | 0.649 | 0.650 | 0.650 | 0.654 | 0.99–1.00 | 1   |

**TABLE 11.** The 5G-NIDD authors' pipeline [11] reproduced and moved one step per rung: balanced accuracy on  $\mathcal{D}_B$ . R0 is their protocol. \*RF on the test flows without the benign copies of Table 9. KNN is evaluated at R0 only.

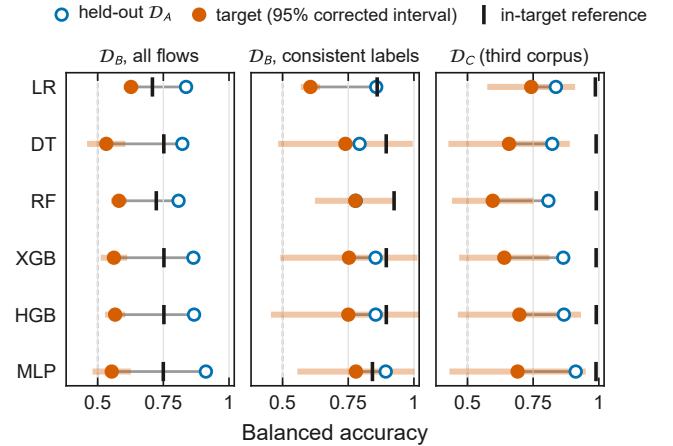
| Rung | Split        | Features      | DT     | Balanced accuracy |        |        |        | RF     |        |
|------|--------------|---------------|--------|-------------------|--------|--------|--------|--------|--------|
|      |              |               |        | RF                | KNN    | NB     | MLP    | BA*    | FPR    |
| R0   | Random 70/30 | Published 10  | 0.9995 | 0.9996            | 0.9988 | 0.9310 | 0.9986 | 0.9997 | 0.0005 |
| R1   | Random 70/30 | No Seq/Offset | 0.7059 | 0.7059            | –      | 0.6814 | 0.7042 | 0.9999 | 0.5881 |
| R2   | Capture file | Published 10  | 0.8594 | 0.9226            | –      | 0.7501 | 0.8669 | 0.9578 | 0.0723 |
| R2   | Capture file | No Seq/Offset | 0.7988 | 0.7996            | –      | 0.7593 | 0.8004 | 0.8958 | 0.1937 |
| R3   | Base station | Published 10  | 0.7619 | 0.8949            | –      | 0.6598 | 0.8577 | 0.9354 | 0.0819 |
| R3   | Base station | No Seq/Offset | 0.6557 | 0.6555            | –      | 0.6271 | 0.6919 | 0.8284 | 0.3465 |

architecture, and averages 0.28 [0.15, 0.41]. The floors do not move. Target ROC-AUC lies between 0.53 and 0.70.

The detectors remain above chance on the target, but not far above it: target balanced accuracy averages 0.57, and for 4 of six architectures its corrected interval excludes 0.5 (3 after Holm correction). For reference, the stratified sampler scores 0.424 macro- $F_1$  on  $\mathcal{D}_B$  and a fair coin would score 0.494; we report both because the value of a macro- $F_1$  floor depends on which prior it samples. LR transfers best by every prevalence-free measure (balanced accuracy 0.63, ROC-AUC 0.70); the MLP, best on held-out source balanced accuracy (0.91), reaches 0.55 on the target. Source performance does not predict target performance in a prevalence-free metric: across the six architectures the rank correlation is -0.26 in balanced accuracy ( $p = 0.62$ ). In macro- $F_1$  it is 0.83 ( $p = 0.04$ ), but macro- $F_1$  ranks move with prevalence, and six architectures give neither value much precision.

*The target's labels.* Part of this loss is the label conflict of Section VI-C. Detectors that learned a UDP flood on  $\mathcal{D}_A$  detect the floods of capture file 15 and, by the same rule, flag between 0.00 and 0.85 of the benign-labeled copies in file 5; LR detects neither. Evaluated without the copies, on a deterministic re-run of 10 of the 20 split seeds, target balanced accuracy is 0.61 to 0.78: 0.74 to 0.78 for the five nonlinear architectures and 0.61 for LR. The loss from held-out source data shrinks to 0.00 to 0.25 and is significant for 1 of six architectures after correction. The conflicts reverse which detector looks transferable: on all flows LR looks best, on flows whose label the record determines it looks worst. Against the target's own reference on those flows (Table 10), transfer still falls short by 0.06 to 0.25 of balanced accuracy.

The reverse direction (Table 13) behaves differently. Balanced accuracy on  $\mathcal{D}_A$  reaches 0.61 to 0.76, and the change



**FIGURE 6.** Balanced accuracy of detectors trained on  $\mathcal{D}_A$ : on held-out  $\mathcal{D}_A$  (open) and on the target (filled, with the Nadeau-Bengio corrected 95% interval), for  $\mathcal{D}_B$  on all flows (20 split seeds),  $\mathcal{D}_B$  without the benign copies of Fig. 5 (10 seeds), and the third corpus  $\mathcal{D}_C$  (20 seeds). The vertical bar is a reference trained and tested inside the target on a random split. The dashed line is chance, where both input-blind baselines sit. On every target the transferred detectors fall short of the in-target reference.

from the (optimistic, random-split) source reference ranges from -0.037 to +0.145: LR, XGB, HGB, and the MLP lose little or nothing, while DT and RF lose significantly. Macro- $F_1$  nonetheless falls sharply, because  $\mathcal{D}_A$  is 94.63% attack and a few false negatives swamp the small benign class. A model trained on the narrower  $\mathcal{D}_B$  thus separates the classes of  $\mathcal{D}_A$  better than a model trained on  $\mathcal{D}_A$  separates those of  $\mathcal{D}_B$ . One possible explanation, which we have not tested, is that  $\mathcal{D}_B$ 's larger and more varied benign traffic gives a better model of benign behavior.

Per-category detection (Table 14) shows that no architec-

**TABLE 12.** Transfer from  $\mathcal{D}_A$  to  $\mathcal{D}_B$  in the 18-column shared space, mean over 20 split seeds.  $\Delta_{BA}$ : source minus target balanced accuracy, with Nadeau–Bengio corrected 95% interval; \*: significant after Holm correction. Last column: macro- $F_1$  loss minus the majority baseline’s loss. Italic rows are the trivial baselines.

| Model         | Macro- $F_1$ |       | Balanced accuracy |       | $\Delta_{BA}$ [95% CI] | Target ROC-AUC | $\Delta_{F_1}$ vs. Maj. |
|---------------|--------------|-------|-------------------|-------|------------------------|----------------|-------------------------|
|               | Src.         | Tgt.  | Src.              | Tgt.  |                        |                |                         |
| LR            | 0.626        | 0.530 | 0.837             | 0.627 | +0.210* [+0.10, +0.32] | 0.701          | -0.018                  |
| DT            | 0.619        | 0.518 | 0.822             | 0.533 | +0.290* [+0.08, +0.50] | 0.526          | -0.013                  |
| RF            | 0.701        | 0.572 | 0.808             | 0.581 | +0.227 [-0.01, +0.47]  | 0.578          | +0.015                  |
| XGB           | 0.646        | 0.532 | 0.864             | 0.562 | +0.303* [+0.14, +0.47] | 0.572          | -0.001                  |
| HGB           | 0.647        | 0.537 | 0.867             | 0.566 | +0.301* [+0.13, +0.47] | 0.581          | -0.004                  |
| MLP           | 0.655        | 0.531 | 0.912             | 0.553 | +0.359* [+0.22, +0.50] | 0.557          | +0.010                  |
| <i>Strat.</i> | 0.494        | 0.424 | 0.502             | 0.500 | +0.001 [-0.01, +0.01]  | 0.500          | -0.044                  |
| <i>Maj.</i>   | 0.492        | 0.378 | 0.500             | 0.500 | +0.000 [+0.00, +0.00]  | 0.500          | +0.000                  |

**TABLE 13.** Reverse direction,  $\mathcal{D}_B$  to  $\mathcal{D}_A$ . The source reference is a random split of  $\mathcal{D}_B$ , which has no group key, and is therefore optimistic.

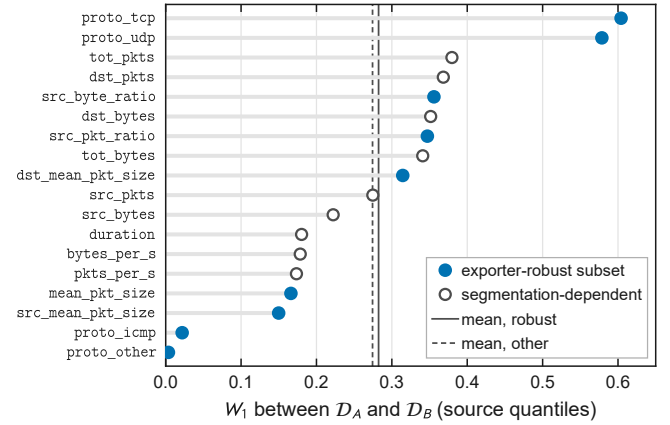
| Model         | Macro- $F_1$ |       | Balanced accuracy |       | $\Delta_{BA}$ [95% CI] | Target ROC-AUC | $\Delta_{F_1}$ vs. Maj. |
|---------------|--------------|-------|-------------------|-------|------------------------|----------------|-------------------------|
|               | Src.         | Tgt.  | Src.              | Tgt.  |                        |                |                         |
| LR            | 0.708        | 0.563 | 0.708             | 0.745 | -0.037* [-0.05, -0.02] | 0.827          | +0.254                  |
| DT            | 0.752        | 0.343 | 0.751             | 0.607 | +0.145* [+0.05, +0.24] | 0.622          | +0.518                  |
| RF            | 0.730        | 0.337 | 0.723             | 0.630 | +0.093* [+0.06, +0.13] | 0.845          | +0.502                  |
| XGB           | 0.753        | 0.497 | 0.752             | 0.764 | -0.012 [-0.03, +0.01]  | 0.845          | +0.364                  |
| HGB           | 0.753        | 0.488 | 0.752             | 0.751 | +0.001 [-0.05, +0.05]  | 0.832          | +0.373                  |
| MLP           | 0.752        | 0.470 | 0.751             | 0.728 | +0.023 [-0.04, +0.09]  | 0.814          | +0.390                  |
| <i>Strat.</i> | 0.502        | 0.417 | 0.502             | 0.500 | +0.003 [+0.00, +0.00]  | 0.500          | +0.194                  |
| <i>Maj.</i>   | 0.377        | 0.486 | 0.500             | 0.500 | +0.000 [+0.00, +0.00]  | 0.500          | +0.000                  |

ture is usable on both classes of  $\mathcal{D}_B$ : LR keeps a benign specificity of 0.98 and detects 0.25 of DoS traffic, while the tree models and the MLP detect more attacks at the price of flagging 0.43 to 0.56 of benign traffic. In the reverse direction, categories absent from  $\mathcal{D}_B$  are not detected uniformly: the ensembles and the MLP recall at most 0.01 of web attacks, whereas brute-force attacks, also absent, are recalled at 0.34 to 0.57.

### E. WHAT THE TRANSFER GAP IS MADE OF

*Separability.* A classifier trained to tell a  $\mathcal{D}_A$  flow from a  $\mathcal{D}_B$  flow reaches 0.999 balanced accuracy on the 18 shared columns and 0.997 on the nine exporter-robust columns alone. The largest per-feature shifts are the protocol mix (most  $\mathcal{D}_A$  flows are TCP and most  $\mathcal{D}_B$  flows UDP), the direction ratios, and the packet counts (Fig. 7), and the robust columns shift as much as the segmentation-dependent ones ( $\bar{W}_1$  of 0.28 against 0.27). So the corpora differ in traffic composition, which no choice of exporter would remove, as well as in flow segmentation, which a common exporter would.

*Model families and adaptation.* Table 15 evaluates the same splits under three changes, and none produces a transferable detector. Restricting the detectors to the nine exporter-robust columns raises held-out source balanced accuracy to 0.80 to 0.94, yet target balanced accuracy falls to 0.48 to 0.53, no better than chance: the columns that do not depend on flow segmentation carry the traffic-composition shift measured above. CORAL, which reads unlabeled target features, gives target balanced accuracy of 0.47 to 0.60; only the MLP gains (0.60 against 0.55 without adaptation), and the target false



**FIGURE 7.** Covariate shift between  $\mathcal{D}_A$  and  $\mathcal{D}_B$  per shared column: first-order Wasserstein distance after the quantile transform fitted on source training data, Eq. (2). Filled markers are the exporter-robust subset, fixed before any result under it was computed; vertical lines are the two group means. The robust columns shift as much as the others, so the gap is not only a matter of how the two exporters segment flows.

positive rate rises to 0.55 to 0.99. The benign-only novelty detectors fail already on held-out source data: in this flow space, floods and scans resemble short benign flows.

*Cost of the shared space.* Projecting  $\mathcal{D}_A$  onto the 18 shared columns costs between 0.005 and 0.082 of in-distribution balanced accuracy relative to all 46 non-port Zeek features (Table 16). The low source macro- $F_1$  of Table 12 reflects the 94.63% attack prevalence of  $\mathcal{D}_A$  rather than weak detectors; their source balanced accuracy is 0.81 to 0.91.

### F. A THIRD CORPUS

To see whether these results depend on the particular pair of corpora, we added  $\mathcal{D}_C$ , the flow records of a 5G core testbed [22], as a second target. It changes the site, the flow exporter, and the vantage point at once, and unlike  $\mathcal{D}_B$  its labels are consistent: only 3.2% of its flows sit on records that also carry the opposite label, and the balanced-accuracy ceiling in the shared columns is 0.992. Trained and tested on  $\mathcal{D}_C$  itself, the six architectures reach 0.987 to 0.990 on a random split, and 0.660 to 0.826 when the capture file of one attack type is held out, so that the attack is unseen.

Transferred from  $\mathcal{D}_A$  (Table 17, 20 split seeds), balanced

**TABLE 14.** Per-category detection, mean over 20 split seeds at  $\tau = 0.5$ . “Ben.” is specificity; other columns are recall. BF: brute force. Categories absent from  $\mathcal{D}_B$  can be tested only in the reverse direction.

| Model | $\mathcal{D}_A \rightarrow \mathcal{D}_B$ |      |       | $\mathcal{D}_B \rightarrow \mathcal{D}_A$ (absent from $\mathcal{D}_B$ : DDoS, BF, web) |      |       |      |      |      |
|-------|-------------------------------------------|------|-------|-----------------------------------------------------------------------------------------|------|-------|------|------|------|
|       | Ben.                                      | DoS  | Probe | Ben.                                                                                    | DoS  | Probe | DDoS | BF   | Web  |
| LR    | 0.98                                      | 0.25 | 0.64  | 0.71                                                                                    | 0.85 | 0.92  | 0.96 | 0.57 | 0.30 |
| DT    | 0.44                                      | 0.65 | 0.29  | 0.82                                                                                    | 0.44 | 0.47  | 0.48 | 0.43 | 0.13 |
| RF    | 0.46                                      | 0.74 | 0.19  | 0.89                                                                                    | 0.41 | 0.55  | 0.49 | 0.34 | 0.00 |
| XGB   | 0.57                                      | 0.58 | 0.17  | 0.89                                                                                    | 0.73 | 0.88  | 0.85 | 0.35 | 0.01 |
| HGB   | 0.54                                      | 0.63 | 0.18  | 0.87                                                                                    | 0.72 | 0.87  | 0.83 | 0.34 | 0.01 |
| MLP   | 0.46                                      | 0.68 | 0.25  | 0.86                                                                                    | 0.71 | 0.73  | 0.81 | 0.36 | 0.00 |

**TABLE 15.** Transfer from  $\mathcal{D}_A$  to  $\mathcal{D}_B$  under alternative feature sets, adaptation, and model families, mean over 20 split seeds. CORAL reads unlabeled target features.

| Arm                           | Model | Src. BA | Tgt. BA | Tgt. AUC | Tgt. FPR |
|-------------------------------|-------|---------|---------|----------|----------|
| Shared space (18)             | LR    | 0.837   | 0.627   | 0.701    | 0.023    |
|                               | DT    | 0.822   | 0.533   | 0.526    | 0.558    |
|                               | RF    | 0.808   | 0.581   | 0.578    | 0.541    |
|                               | XGB   | 0.864   | 0.562   | 0.572    | 0.429    |
|                               | HGB   | 0.867   | 0.566   | 0.581    | 0.463    |
|                               | MLP   | 0.912   | 0.553   | 0.557    | 0.539    |
| Exporter-robust subset (9)    | LR    | 0.804   | 0.510   | 0.583    | 0.854    |
|                               | DT    | 0.883   | 0.480   | 0.488    | 0.177    |
|                               | RF    | 0.894   | 0.507   | 0.469    | 0.206    |
|                               | XGB   | 0.925   | 0.511   | 0.533    | 0.089    |
|                               | HGB   | 0.939   | 0.521   | 0.541    | 0.090    |
|                               | MLP   | 0.928   | 0.534   | 0.499    | 0.039    |
| CORAL (reads target features) | LR    | 0.836   | 0.503   | 0.646    | 0.902    |
|                               | DT    | 0.812   | 0.500   | 0.504    | 0.870    |
|                               | RF    | 0.795   | 0.481   | 0.515    | 0.992    |
|                               | XGB   | 0.906   | 0.473   | 0.478    | 0.906    |
|                               | HGB   | 0.875   | 0.481   | 0.484    | 0.915    |
|                               | MLP   | 0.930   | 0.602   | 0.632    | 0.552    |
| Benign-only novelty           | IF    | 0.503   | 0.492   | 0.464    | 0.024    |
|                               | AE    | 0.455   | 0.479   | 0.484    | 0.074    |

**TABLE 16.** In-distribution cost of the shared space on  $\mathcal{D}_A$  flows: all 46 non-port Zeek features against the 18 shared columns, same splits. Loss interval: Nadeau–Bengio corrected, 95%.

| Model | BA full | BA shared | Loss [95% CI]         | $F_1$ full | $F_1$ shared |
|-------|---------|-----------|-----------------------|------------|--------------|
| LR    | 0.918   | 0.837     | +0.082 [-0.04, +0.20] | 0.725      | 0.626        |
| DT    | 0.861   | 0.822     | +0.039 [-0.12, +0.20] | 0.663      | 0.619        |
| RF    | 0.846   | 0.808     | +0.038 [-0.03, +0.11] | 0.726      | 0.701        |
| XGB   | 0.885   | 0.864     | +0.020 [-0.05, +0.09] | 0.677      | 0.646        |
| HGB   | 0.893   | 0.867     | +0.026 [-0.11, +0.16] | 0.680      | 0.647        |
| MLP   | 0.917   | 0.912     | +0.005 [-0.08, +0.10] | 0.708      | 0.655        |

accuracy on  $\mathcal{D}_C$  is 0.60 to 0.74 and ROC-AUC 0.70 to 0.93. Every architecture detects the SYN flood, a kind of attack it was trained on; recall on the ICMP flood ranges from 0.10 to 1.00, and on the control-plane PFCP attack from 0.00 to 0.99. What fails is the new site’s normal traffic: the false positive rate is 0.225 to 0.807, and up to 0.81 of the core’s own signalling is flagged. The loss from held-out source data, 0.09 to 0.22, varies so much between split seeds that none is significant after correction. Transferred from  $\mathcal{D}_B$  (10 split seeds), balanced accuracy spreads from 0.42 to 0.85, with ROC-AUC as low as 0.38: which architecture transfers depends on the pair of corpora, and none does so consistently. At  $\pi = 0.002$  the best operational precision at any threshold

keeping recall above 10% is 0.092 from  $\mathcal{D}_A$  and 0.031 from  $\mathcal{D}_B$ , and no architecture passes the deployability test in either direction (0 and 0 of six). On a third site, with a third exporter and consistent labels, the pattern of Section VI-D recurs: the SYN flood is found, and the site’s normal traffic is not recognized as normal.

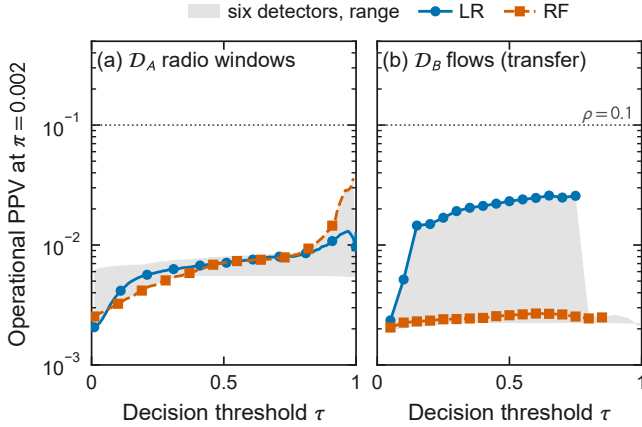
### G. ALERT BURDEN

Table 18 gives pooled operating points at  $\tau = 0.5$  on three evaluation sets, with 95% cluster-bootstrap intervals. On the radio layer, the six detectors have a pooled window false positive rate of 0.22 to 0.33, or 50 to 74 false alerts per benign UE-hour, and at  $\pi = 0.002$  an operational precision of 0.0055 to 0.0080 against a corpus precision of 0.90 to 0.93. These point estimates are not stable. The false positive rate is set by which of the ten benign sessions land in the test set (Table 7), and the bootstrap interval of the pooled rate runs from 0.01 to 0.65 across architectures, or 3 to 146 false alerts per benign UE-hour; operational precision at  $\tau = 0.5$  runs from 0.0028 to 0.13. Thirty sessions do not determine the radio layer’s alert burden; they show only that it depends on the benign conditions a detector meets. On held-out  $\mathcal{D}_A$  flows the false positive rate is 0.066 to 0.201, which at  $\lambda_b = 240,000$  benign flows per hour is 15,726 to 48,289 false alerts per hour; on  $\mathcal{D}_B$  it is 0.023 to 0.558, or 5,620 to 133,990 per hour. Without the benign copies of Section VI-C it is 0.026 to 0.183, still 6,158 to 44,003 false alerts per hour, and operational precision is 0.0072 to 0.0398.

Raising the threshold helps only on held-out  $\mathcal{D}_A$  flows (Fig. 8). The best threshold is chosen on the evaluation data, so the values below are upper bounds. With recall held above 10%, high thresholds reach an operational precision of 0.42 there, but the best any detector reaches on  $\mathcal{D}_B$  is 0.026, and 0.063 without the copies. On the radio layer the best pooled value is 0.036, but its bootstrap interval reaches 1.0 for 4 of six architectures: a resample that omits the benign sessions flagged almost entirely leaves thresholds without a single false positive. The step from corpus to operational precision is arithmetic in  $\pi$ : sweeping  $\pi$  from  $10^{-4}$  to 0.5 moves the collapse factor from 2,627 to 1.18 (Fig. 9), so an operational precision means nothing without the prevalence it assumes. *How operational precision should be estimated.* PPV is strongly non-linear in the false positive rate near zero, so a fold that happens to contain no false positive

**TABLE 17.** Transfer to the third corpus  $\mathcal{D}_C$  (5G core testbed, NFStream flows [22]): balanced accuracy on held-out source data and on  $\mathcal{D}_C$ , target ROC-AUC, recall per attack type, and the share of the core's background signalling flagged (Bkg.). Mean over 20 split seeds from  $\mathcal{D}_A$  and 10 from  $\mathcal{D}_B$ , at  $\tau = 0.5$ .

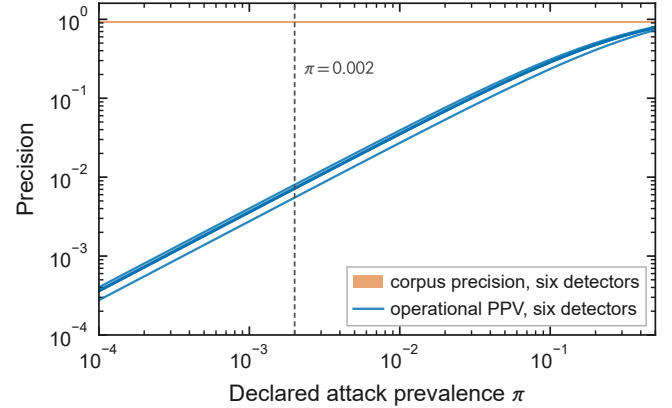
| Model | Src. BA | Tgt. BA | Trained on $\mathcal{D}_A$ |      |      |      |      | Trained on $\mathcal{D}_B$ |       |
|-------|---------|---------|----------------------------|------|------|------|------|----------------------------|-------|
|       |         |         | AUC                        | SYN  | ICMP | PFCP | Bkg. | Tgt. BA                    | AUC   |
| LR    | 0.837   | 0.742   | 0.800                      | 1.00 | 0.10 | 0.00 | 0.27 | 0.517                      | 0.726 |
| DT    | 0.822   | 0.658   | 0.696                      | 1.00 | 1.00 | 0.79 | 0.67 | 0.422                      | 0.384 |
| RF    | 0.808   | 0.596   | 0.926                      | 1.00 | 1.00 | 0.99 | 0.81 | 0.540                      | 0.697 |
| XGB   | 0.864   | 0.640   | 0.888                      | 1.00 | 1.00 | 0.74 | 0.70 | 0.688                      | 0.814 |
| HGB   | 0.867   | 0.698   | 0.932                      | 1.00 | 1.00 | 0.80 | 0.61 | 0.565                      | 0.632 |
| MLP   | 0.912   | 0.691   | 0.837                      | 1.00 | 0.20 | 0.61 | 0.44 | 0.848                      | 0.766 |
| Maj.  | 0.500   | 0.500   | 0.500                      | 1.00 | 1.00 | 1.00 | 1.00 | 0.500                      | 0.500 |

**FIGURE 8.** Pooled operational precision at  $\pi = 0.002$  against the decision threshold on (a) the  $\mathcal{D}_A$  radio layer and (b)  $\mathcal{D}_B$  flows after transfer, at thresholds that keep recall at or above 0.10. The gray band spans the six architectures; the dotted line is the precision bound  $\rho = 0.1$  of the deployability test. No threshold reaches it on either evaluation set.

yields  $PPV = 1$  whatever its recall. Averaging per-fold PPV rewards the architecture that draws the most such folds. On the radio layer the same counts give a spread between the best and worst architecture of 1.45 when pooled, 1.89 as a median over folds, and 16.0 as a mean over folds (Table 19). We report pooled counts throughout.

#### H. CALIBRATION AND PRIOR SHIFT

Under transfer the scores are poorly calibrated (Fig. 10). We fitted three calibrators on a group-disjoint source calibration fold (Table 20). Whether a calibrator can change the achievable operating points depends on whether it preserves the order of scores. Temperature scaling does, and preserves ROC-AUC to within  $2e-05$ . Platt scaling is monotone but its fitted slope can be negative; for DT in 1 of 10 seeds it inverted the ranking and changed ROC-AUC by 0.95. In that seed the tree's scores carried no ranking information on the calibration fold (ROC-AUC 0.48 on 96 held-out source addresses) while they ranked the test fold well (0.97), so Platt scaling fitted the fold faithfully with a slope of -0.55 and inverted the test ranking. The failure belongs to small group-disjoint calibration folds, which need not resemble the test fold, rather than to Platt scaling. Isotonic regression is monotone but piecewise constant: it merges scores into ties

**FIGURE 9.** Pooled operational precision at  $\tau = 0.5$  against the declared attack prevalence  $\pi$  for the six radio-layer detectors (one line each), with their corpus precision as a band. The dashed rule marks  $\pi = 0.002$ . Operational precision follows from the same confusion counts by Eq. (4); only the declared prevalence changes.

and, on its fitting data, yields the ROC convex hull [52]; its largest change in ROC-AUC ranged from 0.04 to 0.21 across architectures. None of the calibrators raises target macro- $F_1$  above the raw scores for any architecture: fitted on a source with 94.63% attacks, they move scores toward that prior and flag more benign traffic on  $\mathcal{D}_B$ .

The obvious remedy for prior shift is to rescale scores to the target prior [48], which a deployment does not know. Two estimators recover it from unlabeled data under the assumption that only the class balance changes: expectation maximization [48] and black-box shift estimation [49]. Here that assumption fails. Against a true target prior of 0.607, their per-architecture mean estimates range from 0.16 to 0.86, and 41% of individual estimates fall below 0.05 or above 0.95. Rescaling with the true prior, an oracle, does not beat the raw scores either (Table 20). The limit is discriminability on the target, with ROC-AUC of 0.53 to 0.70, not calibration.

#### I. DECISION LATENCY

Table 21 reports per-call latency on one host for three implementations of each model: scikit-learn called with a one-row pandas DataFrame, which is how the model is called inside the experiment code; scikit-learn called with a NumPy array on one thread; and ONNX Runtime on one thread, used only where the exported model reproduced scikit-learn's decisions

**TABLE 18.** Pooled operating points at  $\tau = 0.5$ , 20 split seeds. FPR interval: 95% cluster bootstrap resampling split seeds and clusters (radio sessions,  $\mathcal{D}_A$  source addresses,  $\mathcal{D}_B$  capture files; for flows from a deterministic re-run on 10 seeds, EXP-056, on which the “w/o conflicts” block is also computed). “w/o conflicts”:  $\mathcal{D}_B$  without the benign flows whose record also occurs with an attack label. PPV at  $\pi = 0.002$ . <sup>†</sup>False alerts per benign UE-hour (radio, 225 windows per UE-hour) or per hour at  $\lambda_b = 240,000$  benign flows (flows). <sup>‡</sup>Highest pooled PPV over all thresholds with recall at least 0.10.

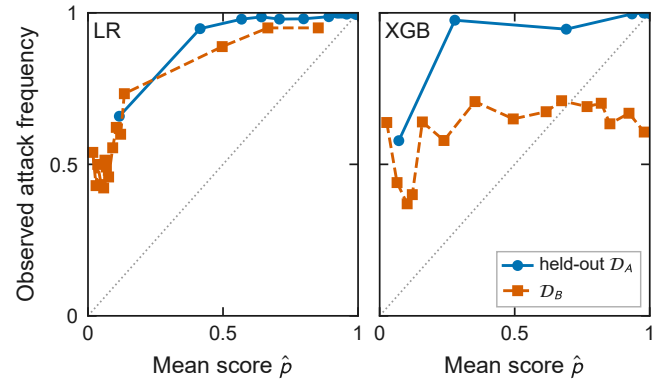
| Corpus                        | Model | TPR   | FPR [95% CI]         | Corpus $P$ | PPV    | False alerts <sup>†</sup> | Best PPV <sup>‡</sup> |
|-------------------------------|-------|-------|----------------------|------------|--------|---------------------------|-----------------------|
| $\mathcal{D}_A$ radio         | LR    | 0.798 | 0.223 [0.017, 0.547] | 0.921      | 0.0071 | 50                        | 0.0130                |
|                               | DT    | 0.910 | 0.329 [0.086, 0.639] | 0.900      | 0.0055 | 74                        | 0.0055                |
|                               | RF    | 0.954 | 0.267 [0.024, 0.650] | 0.921      | 0.0071 | 60                        | 0.0360                |
|                               | XGB   | 0.931 | 0.251 [0.013, 0.606] | 0.924      | 0.0074 | 57                        | 0.0116                |
|                               | HGB   | 0.945 | 0.262 [0.018, 0.615] | 0.922      | 0.0072 | 59                        | 0.0086                |
|                               | MLP   | 0.895 | 0.222 [0.019, 0.560] | 0.929      | 0.0080 | 50                        | 0.0109                |
|                               | Maj.  | 1.000 | 1.000 [1.000, 1.000] | 0.766      | 0.0020 | 225                       | 0.0020                |
| $\mathcal{D}_A$ network       | LR    | 0.892 | 0.201 [0.126, 0.445] | 0.992      | 0.0088 | 48 289                    | 0.0297                |
|                               | DT    | 0.852 | 0.157 [0.068, 0.498] | 0.994      | 0.0108 | 37 658                    | 0.0120                |
|                               | RF    | 0.920 | 0.198 [0.121, 0.611] | 0.993      | 0.0092 | 47 568                    | 0.4172                |
|                               | XGB   | 0.884 | 0.090 [0.027, 0.373] | 0.997      | 0.0193 | 21 625                    | 0.2735                |
|                               | HGB   | 0.881 | 0.066 [0.031, 0.112] | 0.998      | 0.0262 | 15 726                    | 0.2727                |
|                               | MLP   | 0.905 | 0.071 [0.043, 0.349] | 0.997      | 0.0250 | 16 996                    | 0.2189                |
|                               | Maj.  | 1.000 | 1.000 [1.000, 1.000] | 0.967      | 0.0020 | 240 000                   | 0.0020                |
| $\mathcal{D}_B$ (transfer)    | LR    | 0.277 | 0.023 [0.009, 0.079] | 0.948      | 0.0232 | 5 620                     | 0.0258                |
|                               | DT    | 0.624 | 0.558 [0.111, 0.783] | 0.633      | 0.0022 | 133 990                   | 0.0022                |
|                               | RF    | 0.703 | 0.541 [0.121, 0.796] | 0.667      | 0.0026 | 129 843                   | 0.0027                |
|                               | XGB   | 0.552 | 0.429 [0.013, 0.687] | 0.666      | 0.0026 | 102 890                   | 0.0026                |
|                               | HGB   | 0.596 | 0.463 [0.031, 0.699] | 0.665      | 0.0026 | 111 104                   | 0.0026                |
|                               | MLP   | 0.646 | 0.539 [0.030, 0.745] | 0.649      | 0.0024 | 129 320                   | 0.0026                |
|                               | Maj.  | 1.000 | 1.000 [1.000, 1.000] | 0.607      | 0.0020 | 240 000                   | 0.0020                |
| $\mathcal{D}_B$ w/o conflicts | LR    | 0.269 | 0.057 [0.038, 0.086] | 0.947      | 0.0094 | 13 629                    | 0.0102                |
|                               | DT    | 0.662 | 0.183 [0.103, 0.294] | 0.931      | 0.0072 | 44 003                    | 0.0077                |
|                               | RF    | 0.719 | 0.165 [0.118, 0.258] | 0.943      | 0.0087 | 39 561                    | 0.0629                |
|                               | XGB   | 0.531 | 0.026 [0.012, 0.047] | 0.987      | 0.0398 | 6 158                     | 0.0631                |
|                               | HGB   | 0.548 | 0.047 [0.030, 0.080] | 0.978      | 0.0227 | 11 315                    | 0.0587                |
|                               | MLP   | 0.612 | 0.053 [0.026, 0.102] | 0.977      | 0.0225 | 12 794                    | 0.0306                |
|                               | Maj.  | 1.000 | 1.000 [1.000, 1.000] | 0.790      | 0.0020 | 240 000                   | 0.0020                |

**TABLE 19.** Operational precision on the radio layer at  $\tau = 0.5$ ,  $\pi = 0.002$ , estimated three ways from the same 20 folds.

| Model | Folds with FP = 0 | Pooled | Median | Mean   |
|-------|-------------------|--------|--------|--------|
| LR    | 1                 | 0.0071 | 0.0118 | 0.0815 |
| DT    | 0                 | 0.0055 | 0.0070 | 0.0137 |
| RF    | 1                 | 0.0071 | 0.0092 | 0.0814 |
| XGB   | 4                 | 0.0074 | 0.0095 | 0.2185 |
| HGB   | 2                 | 0.0072 | 0.0091 | 0.1291 |
| MLP   | 0                 | 0.0080 | 0.0132 | 0.0476 |

(12 of 12 exports, with decision agreement of at least 1.0000). A no-op call has a 99th percentile of 0.001 ms, so the platform floor does not affect the comparisons.

The implementation moves latency more than the architecture does (Fig. 11). For LR, the DataFrame path costs 0.44 ms at the median against 0.11 ms for an array, the difference being input validation. RF predicting one row through its default thread pool takes 29 ms at the median, 5.6 ms on one thread and 0.012 ms in ONNX Runtime. Across architectures, the 99th percentile of ONNX Runtime lies between 0.014 and 0.035 ms on the radio layer, against up to 21.5 ms for scikit-learn with arrays. A ranking of architectures by latency is really a ranking of implementations, and a comparison with measurements taken on other hosts or in other languages [21] says nothing about the architectures. For context only, Obi-



**FIGURE 10.** Reliability diagrams for LR and XGB on held-out  $\mathcal{D}_A$  and on  $\mathcal{D}_B$ , fifteen equal-mass bins pooled over 20 split seeds.

uwevwi et al. measure inference inside an OpenAirInterface and FlexRIC near-real-time RIC in the microsecond range for logistic regression and a small MLP [21], within an order of magnitude of our ONNX Runtime figures.

For a window-level detector the decision path is window aggregation followed by inference. Aggregating one 16-record window costs 0.064 ms at the 99th percentile, and the upper bound of the 99th percentile of aggregation plus ONNX inference is at most 0.12 ms across the architectures with a verified export. On this host that path would meet any budget

**TABLE 20.** Calibration over 10 split seeds. Left: largest absolute change in ROC-AUC caused by each calibrator, over seeds and both evaluation sets. Right: target macro- $F_1$  at  $\tau = 0.5$  for raw scores, Platt and isotonic calibration, and Platt followed by prior correction with the EM estimate or the true (oracle) target prior.

| Model | max $ \Delta \text{ROC-AUC} $ |                    |       | Target macro- $F_1$ at $\tau = 0.5$ |       |       |       |         |
|-------|-------------------------------|--------------------|-------|-------------------------------------|-------|-------|-------|---------|
|       | Temp.                         | Platt              | Iso.  | Raw                                 | Platt | Iso.  | +EM   | +Oracle |
| LR    | $< 10^{-6}$                   | $1 \times 10^{-6}$ | 0.209 | 0.538                               | 0.415 | 0.391 | 0.327 | 0.478   |
| DT    | $2 \times 10^{-5}$            | 0.946              | 0.170 | 0.538                               | 0.395 | 0.405 | 0.447 | 0.436   |
| RF    | $< 10^{-6}$                   | $< 10^{-6}$        | 0.050 | 0.579                               | 0.504 | 0.470 | 0.440 | 0.488   |
| XGB   | $< 10^{-6}$                   | $< 10^{-6}$        | 0.150 | 0.555                               | 0.450 | 0.394 | 0.456 | 0.518   |
| HGB   | $< 10^{-6}$                   | $< 10^{-6}$        | 0.095 | 0.549                               | 0.429 | 0.399 | 0.479 | 0.507   |
| MLP   | $2 \times 10^{-5}$            | $2 \times 10^{-5}$ | 0.045 | 0.578                               | 0.405 | 0.408 | 0.475 | 0.534   |

in the specified range. A repeat of the radio measurement on the same host, which kept every call for Fig. 11, gave medians 1.02 to 2.1 times those of Table 21 (median ratio 1.46), so per-call latency on a host without CPU isolation depends on the state of the host as well as on the model; the upper 95% bound of the 99th percentile of aggregation plus ONNX inference was still at most 0.36 ms. Whether a deployment does is decided by the terms we did not measure: indication decoding, control encoding, and queueing inside a RIC.

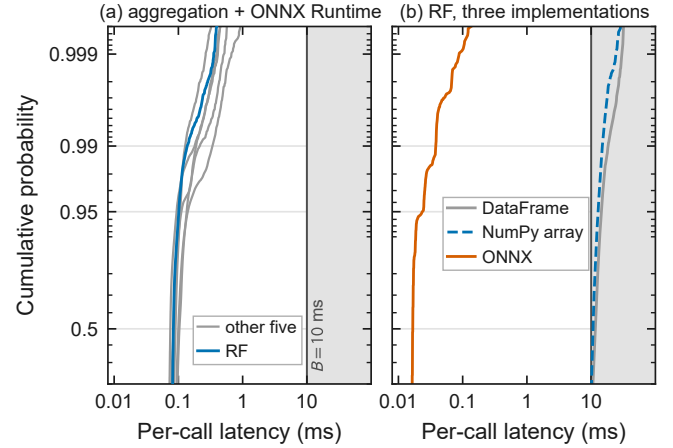
For a flow-level detector, features cost more than inference. On the three DDoS captures of  $\mathcal{D}_A$ , the vectorized exporter produces flow records at 0.20 to 0.90 ms per flow, 22.8 to 41.6 times faster than the per-packet reference implementation on the same captures (4.49 to 37.34 ms per flow); per-flow cost grows with packets per flow. On the 2 captures compared record by record, the two exporters agree exactly in packets and bytes on all 38,032 shared IPv4 flow keys; the reference additionally emits 22 and 41 IPv6 link-local keys, which the vectorized exporter, being IPv4-only, does not parse. Mapping one exported record onto the shared space through the pandas code of our pipeline costs a further 8.5 ms at the median, while ONNX inference on flow features costs at most 0.037 ms at the 99th percentile. These figures describe a Python pipeline on one host: they locate the cost in feature construction, and say nothing about what a native exporter at the user plane would cost.

## VII. DISCUSSION

This section draws the consequences for O-RAN practice, applies the deployability test, and relates the results to the reported state of the art.

### A. IMPLICATIONS FOR O-RAN PRACTICE

**Data path.** E2SM-KPM delivers aggregated measurements per reporting period; it does not carry packets. A window-level detector can run inside the RIC on KPM reports, but a flow-level detector needs a flow exporter next to the user plane, in the central unit or at the user-plane function, and a transport of its records to the xApp, for example through a custom E2 service model or the non-real-time data collection path. That transport adds to  $t_{\text{ind}}$  in Eq. (5), which we did not measure.



**FIGURE 11.** Empirical distribution of per-call latency for the radio layer on one host, 20,000 calls per stage, on a logit scale so that the median, 95th and 99th percentiles are all visible. (a) The decision path of the deployability test, window aggregation followed by ONNX Runtime, for the six architectures. (b) One architecture, RF, under the three implementations of Table 21. The shaded region lies beyond  $B = 10$  ms, the smallest budget in the specified range. Only feature construction and inference are timed; indication decoding, control encoding, and queueing, which need a RIC in the path, are not. The curves come from a repeat of the measurement that kept every call (EXP-060); Table 21 reports the original run.

**Detection delay.** Compute latency is a small part of the time to detect. A window-level decision cannot precede the end of its window (16 s here), and a flow-level decision cannot precede the export of the flow record, which for a long connection waits for an idle or active timeout. The KPM reporting period and the flow timeouts should be reported beside compute latency.

**Acting on alerts.** An alert that drives an E2SM-RC action, such as isolating a UE or steering it, acts on a false alert as readily as on a true one. At the false positive rates of Table 18, automatic mitigation on the radio layer would act against each benign UE 50 to 74 times per hour; alerts of this quality can inform an operator or a non-real-time policy through A1, but should not trigger automatic control.

**Admission and monitoring.** The deployability test of Eq. (6) can serve as an admission check before an xApp is enabled on a new site, provided that site supplies labeled benign and attack traffic. After admission, unlabeled shift detection [53] can flag when the incoming traffic has moved away from the

**TABLE 21.** Per-call latency in milliseconds on one host (Section V): 20,000 calls after 1,000 warm-up calls, one thread per library, garbage collection enabled. Emulated: no RIC in the path. Radio path  $\bar{p}_{99}$ : upper 95% bound of the 99th percentile of window aggregation plus ONNX inference. Dashes: no export that reproduced scikit-learn's decisions.

| Model | sklearn, DataFrame |          | sklearn, array |          | ONNX Runtime |          | Radio path     | Flow layer, $p_{99}$ |       |
|-------|--------------------|----------|----------------|----------|--------------|----------|----------------|----------------------|-------|
|       | $p_{50}$           | $p_{99}$ | $p_{50}$       | $p_{99}$ | $p_{50}$     | $p_{99}$ | $\bar{p}_{99}$ | array                | ONNX  |
| LR    | 0.44               | 0.72     | 0.11           | 0.17     | 0.010        | 0.023    | 0.12           | 0.16                 | 0.025 |
| DT    | 0.31               | 0.52     | 0.03           | 0.04     | 0.010        | 0.014    | 0.11           | 0.06                 | 0.024 |
| RF    | 6.25               | 12.99    | 5.57           | 21.50    | 0.012        | 0.024    | 0.10           | 4.45                 | 0.029 |
| XGB   | 0.73               | 1.00     | 0.16           | 0.32     | 0.011        | 0.021    | 0.09           | 0.40                 | 0.024 |
| HGB   | 1.48               | 2.72     | 1.17           | 1.92     | 0.013        | 0.035    | 0.12           | 1.22                 | 0.034 |
| MLP   | 0.43               | 0.67     | 0.11           | 0.21     | 0.017        | 0.035    | 0.11           | 0.15                 | 0.037 |

training distribution; the corpora here are separable at 0.999 balanced accuracy, so such a monitor would fire immediately.

### B. APPLYING THE DEPLOYABILITY TEST

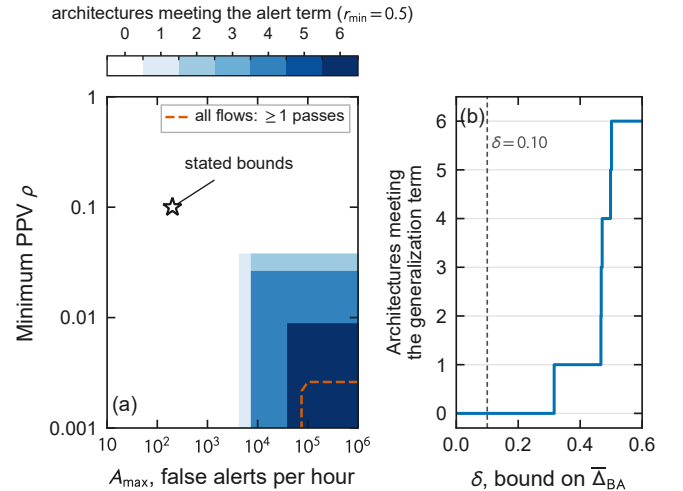
We instantiate Eq. (6) with the following bounds. They are our choice: no O-RAN specification fixes them, and an operator would set them from its own traffic and staffing. The generalization bound is  $\delta = 0.10$  of balanced accuracy, applied to the upper end of the corrected 95% interval. The alert term asks for a threshold that detects at least half of the attacks ( $r_{\min} = 0.5$ ), makes at least one alert in ten a true one ( $\rho = 0.1$  at  $\pi = 0.002$ ), and raises at most  $A_{\max} = 200$  false alerts per hour at  $\lambda_b = 240,000$  benign flows per hour. All alert conditions are evaluated on  $\mathcal{D}_B$ , because an operator deploys on the target, not on the source. The latency term uses the strictest budget the specification allows,  $B = 10$  ms: the upper bound of the 99th percentile of the radio decision path meets 1 ms for every architecture (Section VI-I), so the term passes anywhere in the specified range, with the caveat that indication decoding, control encoding, and queueing were not measured.

Table 22 gives the result. No architecture passes: 0 of six. Every architecture fails the generalization term, with upper bounds of 0.32 to 0.50, and every architecture fails the alert term by a wide margin: at recall of at least 0.5, the best operational precision on  $\mathcal{D}_B$  is 0.0027 and the fewest false alerts per hour is 95,507. The verdict does not hinge on the bounds chosen: it holds for any  $\rho$  above 0.0027, any  $A_{\max}$  below 95,507 per hour, and any  $\delta$  below 0.32, and it does not depend on the latency term at all. It also survives the label conflicts of Section VI-C: evaluated on  $\mathcal{D}_B$  without the benign copies, 0 of six pass, the best operational precision at recall of at least 0.5 is 0.0419, and the fewest false alerts per hour are 5,603. Fig. 12 shows how far the bounds would have to move: an architecture meets the alert term only once an operator accepts several thousand false alerts per hour and a precision of a few percent, and meets the generalization term only once  $\delta$  exceeds 0.32. The latency term would matter only once the first two pass; on our off-platform measurements it does not decide the verdict.

*Is the test attainable?* A test that no detector passes could simply be too strict. The published 5G-NIDD pipeline of Section VI-C shows that it is not. On its own random split, with the test's alert term evaluated on that split, 2 of its

**TABLE 22.** Deployability test of Eq. (6) per architecture.  $\bar{\Delta}_{BA}$ : upper 95% bound of the balanced-accuracy loss (pass if  $\leq 0.10$ ). Best PPV and minimum false alerts per hour over thresholds with recall  $\geq 0.5$  on  $\mathcal{D}_B$  (pass if PPV  $\geq 0.1$  and alerts  $\leq 200$ ).  $B_{\min}$ : smallest budget met by the upper bound of the radio decision path's 99th percentile, timed off-platform.

| Model | $\bar{\Delta}_{BA}$ | Gen. | Best PPV | Min. alerts/h | Alert | $B_{\min}$ (ms) | $\mathcal{P}$ |
|-------|---------------------|------|----------|---------------|-------|-----------------|---------------|
| LR    | 0.32                | ×    | 0.0024   | 166 328       | ×     | 1               | 0             |
| DT    | 0.50                | ×    | 0.0022   | 118 342       | ×     | 1               | 0             |
| RF    | 0.47                | ×    | 0.0027   | 98 410        | ×     | 1               | 0             |
| XGB   | 0.47                | ×    | 0.0026   | 95 507        | ×     | 1               | 0             |
| HGB   | 0.47                | ×    | 0.0026   | 99 277        | ×     | 1               | 0             |
| MLP   | 0.50                | ×    | 0.0026   | 105 744       | ×     | 1               | 0             |



**FIGURE 12.** Sensitivity of the deployability test to its bounds. (a) Number of the six architectures meeting the alert term ( $r_{\min} = 0.5$ ,  $\pi = 0.002$ ,  $\lambda_b = 240,000$  flows per hour) on  $\mathcal{D}_B$  without the benign copies, as the alert bound  $A_{\max}$  and the precision bound  $\rho$  vary; on all flows, at least one architecture passes only below and to the right of the dashed line. The star marks the bounds we use. (b) Number meeting the generalization term as  $\delta$  varies. Both terms must hold.

five classifiers pass: operational precision reaches 0.79 at  $\pi = 0.002$  with as few as 128 false alerts per hour at  $\lambda_b = 240,000$ . The same pipeline fails the test one rung later, when the two position counters are removed, and at every later rung, on all flows and on the flows without the benign copies (Table 11). The test is therefore only as informative as the evaluation it is applied to: applied to a random split that still contains record positions, it certifies the split, and the pass

disappears as soon as the evaluation moves one step toward deployment conditions. The detectors of this paper fail it on the evaluations closest to deployment that the corpora allow.

### C. RELATION TO REPORTED RESULTS

Published O-RAN and 5G intrusion detection results are mostly in-distribution scores on corpora we do not share, and our own in-distribution scores under a random split are high as well (Table 5). A direct numerical comparison would compare protocols, not detectors. What the results above allow is a statement about which reported quantities predict deployment: held-out macro- $F_1$  under a random split does not, and the quantities in the reporting requirements of Section X are those that changed the conclusions here.

## VIII. REPRODUCIBILITY AND ETHICS

Code, configuration, the seeded split procedures from which every split is regenerated deterministically, raw per-seed results, and the generators for every table, figure, and in-text number are available at <https://github.com/adeliusa486/oran-ids-deployability>. One script regenerates every number, table, and figure of this paper from the shipped per-seed results and checks them against the manuscript, and the same script re-runs each experiment from the raw corpora, whose files it verifies by SHA-256; the repository's `REPRODUCE.md` gives the steps. All three corpora are public:  $\mathcal{D}_A$  under CC-BY-4.0 from its authors [39],  $\mathcal{D}_B$  from its authors [37], and  $\mathcal{D}_C$  from its authors' repository [22], which states no license, so we do not redistribute it. The traffic comes from testbed devices and a test network, not from subscribers, so no personal data was processed. Publishing where detectors fail carries a modest dual-use risk; we judge it outweighed by the benefit to operators of knowing that reported accuracy does not transfer.

## IX. LIMITATIONS

*Exporter.*  $\mathcal{D}_A$ ,  $\mathcal{D}_B$ , and  $\mathcal{D}_C$  were exported by three different tools (Zeek, Argus, NFStream), and we could not re-extract  $\mathcal{D}_A$  or  $\mathcal{D}_B$  with a common exporter: our copy of  $\mathcal{D}_B$  contains Argus flow records only, and no Zeek installation was available on the measurement host. The domain classifier and the robust subset show that the corpora differ in traffic composition as well as in flow segmentation, but they do not measure how much of the transfer gap a common exporter would remove. The base-station split inside  $\mathcal{D}_B$  (Section VI-C) changes the site while keeping the exporter, which bounds part of the question from one side.

*Target labels.* Of  $\mathcal{D}_B$ 's benign flows, 59% are exact copies of records labeled as UDP flood elsewhere in the corpus (Section VI-C). We cannot tell from the published files which label is correct, so every  $\mathcal{D}_B$  operating point is reported with and without these flows. The assignment of the two capture passes to the two base stations is inferred from the repeated capture schedule and the restarted row index; the dataset does not state it.

*Radio capture schedule.* Nine of the ten benign radio sessions were recorded six weeks before any attack session, mostly with one UE, and every radio label is a session label. A radio detector may learn to separate capture periods as well as attacks. The one benign session from the attack period is flagged almost entirely, but so are two early sessions (Table 7), and ten benign sessions cannot separate the two explanations.

*Three corpora.* Three corpora establish that the gap recurs, not how it varies across deployments. The third is a 5G core testbed rather than an O-RAN deployment, holds only floods and a control-plane attack, and has three capture files, too few for cluster intervals. No third O-RAN corpus with paired radio telemetry was available to us.

*Latency.* All latency figures are emulated on one Windows host without CPU isolation, and  $t_{\text{ind}}$ ,  $t_{\text{act}}$ , and  $t_q$  are not measured. CPU and memory use under load are not measured.

*Operational parameters and the test.* The prevalence  $\pi = 0.002$  and the benign arrival rate  $\lambda_b$  are declared, not measured on any network; we sweep both, and operational precision depends on  $\pi$  alone. The bounds of the deployability test are our choice (Section VII-B). Fig. 12 gives the verdict under other bounds, but an operator may weigh recall, precision, and alert volume differently, and the test does not price detection delay or the cost of a missed attack.

*Sample size.* The radio layer has 30 capture sessions, each with a single scenario, so session and attack scenario are confounded, and the protocol and temporal analyses can detect large effects only. With ten benign sessions, the radio layer's false positive rate, and with its operational precision, carries a wide interval (Section VI-G). The transfer statistics vary training sets over 20 seeds but evaluate one fixed target corpus, so they do not describe other deployments.

*Scope.* We evaluate binary detection. Multi-class attribution, feature attribution [62], and the fusion of radio and flow features are left open; the published summary files cannot be joined record by record, and fusion requires the raw archives, as in [31]; three attack categories of  $\mathcal{D}_A$  are absent from  $\mathcal{D}_B$  and can be tested in one direction only.

## X. CONCLUSION AND REPORTING REQUIREMENTS

A held-out score on one corpus says little about how a detection xApp behaves elsewhere. On the corpora studied here, a random split inflates the score, and a model-free lookup gains as much, which points to the recognition of capture sessions. The radio corpus's capture schedule leaves no room to estimate drift, and the benign session a detector happens to meet sets its false positive rate. A published 99.9% on the target corpus rests on record-position fields, and half of that corpus carries labels its records cannot determine. On flows with consistent labels, detectors transferred from the other corpus reach balanced accuracy 0.61 to 0.78, below what the target itself allows. On a third corpus they find the SYN flood and flag much of the normal traffic. At a realistic base rate, no threshold gives usable operational precision, and latency, the one term that passes, was timed off-platform

only. We therefore propose that papers claiming a deployable O-RAN detection capability report at least: (i) performance on a corpus collected independently of the training data, in a prevalence-invariant metric such as balanced accuracy or ROC-AUC, beside input-blind baselines and a reference trained inside that corpus; (ii) the split protocol and its group key, results under a group-disjoint split, and the category coverage of any time-based split; (iii) the false positive rate on benign traffic, with its unit, the resulting alert rate, and the operational precision at a stated base rate, all from pooled confusion counts with intervals that resample clusters; (iv) the operating threshold, how it was chosen, and the recall it gives up; (v) intervals that account for overlap between resampled splits; (vi) tail latency with its stages, implementation, and host, measured on a real RIC before any conformance claim is made; and (vii) evidence that the labels are a function of the features used, with no identical records under opposite labels, and that no record-position or identifier field is a feature.

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